MEMO

Paper 1 mock: marking memo

Tick-per-mark answers, the working DBE memos want to see, and the traps that cost marks in real memos.

Marking rules that apply to every question (from the DBE memos)

- **Consistent accuracy (CA):** an error carried from an earlier part is not penalised again. Marking stops at the second calculation error in the same item.
- **Answer only:** full marks only where the memo says so. Otherwise working is needed. If a candidate answers a question twice, only the first attempt is marked.
- **Rounding:** two decimals unless asked. In finance, early rounding that changes the final answer loses the answer mark; money answers to the nearest cent.
- **Sketches:** marks are for shape, intercepts with the axes, turning points and asymptotes, each labelled with coordinates.
- **First principles:** notation matters: the limit must be written on every line until h is gone; dropping "$\lim_{h \to 0}$" too early loses a mark.
- **Inequalities and intervals:** both the critical values AND the correct region are marked; check the notation ($\leq$ vs $<$, round vs square brackets).

What is NOT on the information sheet (learn it)

Algebra

- Nature of roots from $\Delta = b^2 - 4ac$: $\Delta < 0$ non-real; $\Delta = 0$ real, equal; $\Delta > 0$ real, unequal; Δ a perfect square: rational
- Exponent laws: $a^m \cdot a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$, $a^{-n} = \frac{1}{a^n}$, $a^{\frac{m}{n}} = \sqrt[n]{a^m}$
- Square both sides of a surd equation, then CHECK each answer in the original equation
- Quadratic inequality: find the critical values, then sketch or use a number line; $a(x-p)(x-q) \leq 0$ with $a > 0$ gives $p \leq x \leq q$

Patterns, sequences, series

- Quadratic pattern $T_n = an^2 + bn + c$: $2a$ = second difference, $3a + b = T_2 - T_1$, $a + b + c = T_1$
- Arithmetic: constant difference $d = T_2 - T_1$; geometric: constant ratio $r = \frac{T_2}{T_1}$
- Sigma: number of terms $=$ top $-$ bottom $+1$
- A geometric series converges only if $-1 < r < 1$ ($r \neq 0$)

Functions

- Hyperbola $y = \frac{a}{x-p} + q$: asymptotes $x = p$, $y = q$; axes of symmetry $y = x - p + q$ and $y = -x + p + q$
- Parabola $y = a(x-p)^2 + q$: turning point $(p; q)$; axis $x = -\frac{b}{2a}$
- Exponential $y = ab^{x-p} + q$: asymptote $y = q$
- Inverse: swap x and y; $y = b^x \Leftrightarrow x = \log_b y$; the inverse of $y = ax^2$ is a function only if the domain is restricted
- Log laws: $\log_a xy = \log_a x + \log_a y$, $\log_a \frac{x}{y} = \log_a x - \log_a y$, $\log_a x^n = n \log_a x$
- Reflection in y-axis: $f(-x)$; in x-axis: $-f(x)$; in $y = x$: f^{-1}; shift: $f(x-k)$ right, $f(x) + k$ up

Finance, calculus, probability

- Effective rate: $1 + i_{\text{eff}} = \left(1 + \frac{i^{(m)}}{m}\right)^m$
- Outstanding balance = present value of the remaining payments
- $\frac{d}{dx}(ax^n) = anx^{n-1}$; tangent: $m = f'(x)$ at the point
- Stationary points: $f'(x) = 0$; point of inflection: $f''(x) = 0$; concave up: $f''(x) > 0$
- f increasing where $f'(x) > 0$; max/min: $f'(x) = 0$ (optimisation, also check endpoints)
- Independent: $P(A \text{ and } B) = P(A) \times P(B)$; mutually exclusive: $P(A \text{ and } B) = 0$; complementary: $P(\text{not } A) = 1 - P(A)$
- Arrangements of n different objects: $n!$; with k identical: divide by $k!$

QUESTION 1

1.1.1 (2)

$x = 4$ ✓ or $x = -5$ ✓

Accept $2x - 8 = 0$ and $x + 5 = 0$ shown, then the two answers. When one factor is x itself, as in $x(x + 9) = 0$, the answers are $x = 0$ and $x = -9$.

1.1.2 (2)

$x(x - 8) = 0$ ✓

$x = 0$ or $x = 8$ ✓

> **Memo trap:** Dividing both sides by x loses the answer $x = 0$.

1.1.3 (3)

$(x - 4)(x + 7) = 0$ ✓

$x = 4$ ✓ or $x = -7$ ✓

1.1.4 (3)

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(3)(1)}}{2(3)}$$ ✓

$$x = \frac{7 \pm \sqrt{37}}{6}$$

$x = 2{,}18$ ✓ or $x = 0{,}15$ ✓

Calculator: $x = 2{,}1805\ldots$ or $x = 0{,}1529\ldots$. Answer only: full marks. In 2017 the same type carried 4 marks (substitution, simplification, two answers).

1.1.5 (4)

$4x^2 - 7x + 1 = 0$ ✓

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(4)(1)}}{2(4)}$$ ✓

$x = 1{,}59$ ✓ or $x = 0{,}16$ ✓

Answer only: 4 marks. Use the formula only after moving every term to one side.

1.1.6 (3)

$(x - 4)(x + 3) > 0$ ✓

Critical values: $x = 4$ and $x = -3$

$x < -3$ ✓ or $x > 4$ ✓

Sketch the parabola: it is above the x-axis OUTSIDE the roots. Interval form $(-\infty\,;-3) \cup (4\,;\infty)$ is accepted. The word *or* (not *and*) must be used. The 2024 version of this type carried 4 marks (factors, critical values, answer for 2); answer only earned 3 of 4.

1.1.7 (4)

$x^2 - x - 56 > 0$ ✓

$(x - 8)(x + 7) > 0$; critical values $x = 8$ and $x = -7$ ✓

$x < -7$ ✓ or $x > 8$ ✓

> **Memo trap:** Moving x to the left changes its sign: $x^2 - 56 - x > 0$, not $x^2 - 56 + x > 0$.

1.1.8 (4)

$x^2 + 7x + 10 \leq 0$ ✓

$(x + 5)(x + 2) \leq 0$; critical values $x = -5$ and $x = -2$ ✓

$-5 \leq x \leq -2$ ✓✓

Interval form $[-5\,;-2]$ is accepted. The answer is BETWEEN the roots and the end points are included because of $\leq$.

1.1.9 (3)

Critical values: $x = 2$ and $x = -5$ ✓

Sketch / sign table: $(2 - x)(x + 5)$ is a parabola that opens DOWNWARDS, so it is negative outside the roots ✓

$x < -5$ or $x > 2$ ✓

> **Memo trap:** Multiplying out gives $-x^2 - 3x + 10 < 0$. Dividing by -1 reverses the inequality sign: $x^2 + 3x - 10 > 0$.

1.1.10 $(3x-2)(3x+2)<0$ ✓ (3)

Critical values: $x=-\frac{2}{3}$ and $x=\frac{2}{3}$ ✓

$-\frac{2}{3}<x<\frac{2}{3}$ ✓

Also accepted: $x^2<\frac{4}{9}$ followed by $-\frac{2}{3}<x<\frac{2}{3}$.

1.1.11 $6x^2-9x>0$ ✓ (4)

$3x(2x-3)>0$; critical values $x=0$ and $x=\frac{3}{2}$ ✓

$x<0$ ✓ or $x>\frac{3}{2}$ ✓

> **Memo trap:** Dividing by x is NOT allowed: x can be negative, so the solution $x<0$ is lost.

1.1.12 $x+10=(x-2)^2=x^2-4x+4$ ✓ (5)

$x^2-5x-6=0$ ✓

$(x-6)(x+1)=0$ ✓

$x=6$ or $x=-1$ ✓

Check: $x=-1$ gives $\sqrt{9}=3\neq-3$, so $x=-1$ is rejected. $x=6$ ✓

In the 2023 version ($\sqrt{2x+1}=x-1$, 4 marks) the standard form $x^2-4x=0$ is solved directly, so there was no separate mark for factors.

When the right side is of the form $a-x$ (for example $\sqrt{x+7}=5-x$), a root is rejected as soon as $a-x<0$.

> **Memo trap:** Squaring $x-2$ gives x^2-4x+4, NOT x^2+4.

1.1.13 $\sqrt{2x-6}=x-3$ ✓ (5)

$2x-6=(x-3)^2=x^2-6x+9$ ✓

$x^2-8x+15=0$ ✓

$(x-3)(x-5)=0$; $x=3$ or $x=5$ ✓

Check: $x=3$: $\sqrt{0}+3=3$ and $x=5$: $\sqrt{4}+3=5$. Both answers are valid ✓

Alternative: let $k=\sqrt{2x-6}$, so $x=\frac{k^2+6}{2}$ and $k+3=\frac{k^2+6}{2}$ leads to $k=0$ or $k=2$ and the same two answers. When a root IS rejected (for example $\sqrt{5-x}-x=1$ gives $(x+4)(x-1)=0$, and only $x=1$ works), the final mark is for the rejection.

1.1.14 $x^2-12=4x$ ✓ (4)

$x^2-4x-12=0$ ✓

$(x-6)(x+2)=0$ ✓

$x=6$; $x=-2$ is rejected, because $\sqrt{x}$ is not defined for $x<0$ ✓

1.1.15 $4x=(x-3)^2=x^2-6x+9$ ✓ (4)

$x^2-10x+9=0$ ✓

$(x-9)(x-1)=0$ ✓

$x=9$; $x=1$ is rejected ($2\sqrt{1}=2\neq-2$) ✓

> **Memo trap:** $(2\sqrt{x})^2=4x$, NOT $2x$.

The same method is used when the x term and a number are on one side, as in $x-9\sqrt{x}=-14$ (2022). Isolate the root, square, solve, and check BOTH answers: there both work. Substitution $k=\sqrt{x}$ turns it into $k^2-9k+14=0$.

1.1.16 (4)

$\sqrt{50-x^2}=x$ ✓

$50-x^2=x^2$ ✓

$x^2=25$ ✓

$x=5$; $x=-5$ is rejected, because a square root is not negative ✓

The product of the two square roots is the square root of the product: $(\sqrt{50}+x)(\sqrt{50}-x)=50-x^2$.

1.1.17 (5)

Let $k=\dfrac{1}{\sqrt{x}}$; then $\sqrt{\dfrac{1}{x}}=k$ and the equation is $\sqrt{k+6}=k$ ✓

$k+6=k^2$ ✓

$k^2-k-6=0$ ✓

$(k-3)(k+2)=0$; $k=3$ or $k=-2$ ✓

$k=-2$ is rejected ($k>0$) ; $\dfrac{1}{\sqrt{x}}=3$, so $x=\dfrac{1}{9}$ ✓

Without a substitution: square both sides, isolate the remaining surd, square again, then reject the extra root. Same 5 marks.

1.1.18 (4)

$5^x\cdot 5^2-5^x\cdot 5^1=500$ ✓

$5^x(25-5)=500$ ✓

$5^x=25=5^2$ ✓

$x=2$ ✓

Alternative: common factor 5^{x+1}: $5^{x+1}(5-1)=500$, $5^{x+1}=125$, $x+1=3$, $x=2$.

1.1.19 (5)

$3^{2x}-6\cdot 3^x-27=0$ ✓

$(3^x-9)(3^x+3)=0$ ✓

$3^x=9$ ✓

$3^x=-3$ has no solution, because $3^x>0$ ✓

$x=2$ ✓

Substitution $k=3^x$ is fine: $k^2-6k-27=0$. Writing $3^x=-3$ and stating no solution earns its own mark.

1.1.20 (4)

$(4\cdot 2^x-1)(2^x-8)=0$ ✓

$2^x=\dfrac{1}{4}$ or $2^x=8$ ✓

$x=-2$ ✓

$x=3$ ✓

Let $k=2^x$: $4k^2-33k+8=0$. Both values of k are positive, so both give an answer.

1.2.1 (2)

$(x+5)(x-3)=0$ ✓

$x=-5$ or $x=3$ ✓

1.2.2 (2)

$-5<x<3$ ✓✓

Interval form $x\in(-5\,;3)$ is accepted. Both marks are for the final answer. The graph of f is below the x-axis between its roots.

1.2.3 (2)

$f'(x)=2x+2$ ✓

$2x+2\geq 0$, so $x\geq -1$ ✓

The turning point of f is at $x=\dfrac{-5+3}{2}=-1$. The function is increasing to the right of it.

1.3.1 (3)

$(x+5)(x+5)>0$ or $(x+5)^2>0$ ✓

$x\in\mathbb{R}$ ✓, $x\neq -5$ ✓

Accepted: $x<-5$ or $x>-5$. A perfect square is never negative and is zero only at $x=-5$.

1.3.2 $x^2 + 10x + 25 - p = 0$ (4)

Unequal real roots: $\Delta > 0$, so $100 - 4(25 - p) > 0$ ✓, which gives $p > 0$ ✓

Both roots negative: product $25 - p > 0$ (the sum -10 is already negative) ✓

$0 < p < 25$ ✓

Graph method: f has its turning point at $(-5\,;0)$ and cuts the y-axis at $(0\,;25)$. The line $y = p$ cuts the graph twice to the left of the y-axis when $0 < p < 25$. Marks: identify 0 and 25; the correct interval.

1.4.1 $(3y)^2 - 4(3y)y = -27$ ✓ (4)

$-3y^2 = -27$, so $y^2 = 9$ ✓

$y = 3$ or $y = -3$ ✓

$x = 9$ or $x = -9$ ✓

Solutions: $(9\,;3)$ and $(-9\,;-3)$. Alternative: substitute $y = \frac{x}{3}$; $x^2 - \frac{4x^2}{3} = -27$ gives $x^2 = 81$.

1.4.2 $y = 2x - 5$ ✓ (6)

$x^2 - x(2x - 5) - (2x - 5)^2 = -5$ ✓

$-5x^2 + 25x - 20 = 0$, so $x^2 - 5x + 4 = 0$ ✓

$(x - 1)(x - 4) = 0$ ✓

$x = 1$ or $x = 4$ ✓

$y = -3$ or $y = 3$ ✓

Solutions: $(1\,;-3)$ and $(4\,;3)$. Alternative: $x = \frac{y + 5}{2}$ with the same marks. The 2020 and 2024 versions of this type had an xy term and only one squared variable; the 2024 one carried 5 marks (equation 1, substitution, standard form, x-values, y-values).

> **Memo trap:** $(2x - 5)^2 = 4x^2 - 20x + 25$. Forgetting the middle term costs the standard-form mark and everything after it. Always write the substituted expression in brackets, especially after a minus sign.

1.4.3 $y = 2x + 1$ ✓ (6)

$x^2 - 3(2x + 1) = -8$ ✓

$x^2 - 6x + 5 = 0$ ✓

$(x - 1)(x - 5) = 0$ ✓

$x = 1$ or $x = 5$ ✓

$y = 3$ or $y = 11$ ✓

Solutions: $(1\,;3)$ and $(5\,;11)$.

1.4.4 $y = 7 - 2x$ ✓ (5)

$x(7 - 2x) = 6$ ✓

$2x^2 - 7x + 6 = 0$ ✓

$(2x - 3)(x - 2) = 0\,;\, x = \frac{3}{2}$ or $x = 2$ ✓

$y = 4$ or $y = 3$ ✓

Solutions: $(\frac{3}{2}\,;4)$ and $(2\,;3)$. The 2019 version had a product equal to a linear expression ($xy = 14 - 3x$): substitute in the same way, then rearrange to standard form.

1.4.5 (5)

$x = 3y - 3$ ✓

$\frac{3}{3y-3} - \frac{1}{y} = \frac{1}{2}$, so $\frac{1}{y-1} - \frac{1}{y} = \frac{1}{2}$; multiply by $2y(y-1)$: $2y - 2(y-1) = y(y-1)$ ✓

$y^2 - y - 2 = 0$ ✓

$(y-2)(y+1) = 0$; $y = 2$ or $y = -1$ ✓

$x = 3$ or $x = -6$ ✓

Solutions: $(3\,;2)$ and $(-6\,;-1)$. Restrictions: $x \neq 0$, $y \neq 0$ and $y \neq 1$ (neither answer breaks them).

1.4.6 (6)

$x = y + 3$ ✓

$2xy = x^2 + 7$ ✓

$2y(y+3) = (y+3)^2 + 7$ ✓

$2y^2 + 6y = y^2 + 6y + 16$, so $y^2 - 16 = 0$ ✓

$y = 4$ or $y = -4$ ✓

$x = 7$ or $x = -1$ ✓

Solutions: $(7\,;4)$ and $(-1\,;-4)$. Alternative: $y = x - 3$ and $2x(x-3) = x^2 + 7$, $x^2 - 6x - 7 = 0$, $(x-7)(x+1) = 0$.

1.5.1 (4)

$3^m(3^2 - 1) = 2^n(2+1)$ ✓

$3^m \cdot 2^3 = 2^n \cdot 3^1$ ✓

$m = 1$ and $n = 3$ ✓

$m + n = 4$ ✓

Alternative: $3^{m-1} = 2^{n-3}$. A power of 3 can only equal a power of 2 when both are 1, so $m - 1 = 0$ and $n - 3 = 0$.

1.5.2 (4)

$2^3 \cdot 3^{y+1} = (72)^{x/2}$ ✓

$2^3 \cdot 3^{y+1} = 2^{3x/2} \cdot 3^x$ (using $72 = 2^3 \cdot 3^2$) ✓

$\frac{3x}{2} = 3$, so $x = 2$ ✓

$y + 1 = x = 2$, so $y = 1$ ✓

Equate the exponents of each prime base separately. Check: $8 \cdot 3^2 = 72$ and $\sqrt{72^2} = 72$.

1.5.3 (3)

$(n^2)^{30} < (7^3)^{30}$ ✓

$n^2 < 343$ ✓

$n = 18$ ✓

Logs: $n < 7^{1,5} = 18{,}52$, so $n = 18$. Check: $18^2 = 324 < 343$ but $19^2 = 361 > 343$.

1.5.4 (4)

$[2^{x-2}]^4 = 2^{4x-8} = \frac{2^{4x}}{2^8} = \frac{81}{256}$ ✓✓

$\sqrt{7\sqrt{q}} = \sqrt{81} = 9$ ✓

$\frac{81}{256 \times 9} = \frac{9}{256}$ ✓

1.5.5 (3)

$2 \cdot 3^n - 3^n \cdot 3^1 + 3^n \cdot 3^2$ ✓

$= 3^n(2 - 3 + 9)$ ✓

$= 8 \cdot 3^n = 2(4 \cdot 3^n)$, which is 2 times an integer, so it is even ✓

Accepted: any whole number multiplied by an even number is even.

1.5.6 Multiples of 5: 5; 10; 15; 20; 25; 30; 35; 40 ✓ (4)

There are 8 multiples of 5, each giving at least 5^1 ✓

25 is 5^2 and gives ONE extra factor of 5 (the other multiples give only 5^1) ✓

$k = 8 + 1 = 9$ ✓

Alternative: $\left\lfloor \frac{40}{5} \right\rfloor + \left\lfloor \frac{40}{25} \right\rfloor = 8 + 1 = 9$.

1.5.7 $\frac{4}{3} \times \frac{5}{4} \times \frac{6}{5} \times \ldots \times \frac{n}{n-1} \times \frac{n+1}{n}$ ✓ (3)

$= \frac{n+1}{3}$ (all the other numerators and denominators cancel) ✓

$n + 1$ must be a multiple of 3, so $n = 5;\ 8;\ 11;\ 14; \ldots$ (that is $n = 3t - 1$ with $t \geq 2$) ✓

Check: $n = 5$ gives $\frac{4}{3} \times \frac{5}{4} \times \frac{6}{5} = 2$. Pattern method: list the products for $n = 3;\ 4;\ 5;\ 6 \ldots$ and spot the integers.

1.5.8 $\frac{b}{a} = \frac{c}{b}$ ✓ (4)

$b^2 = ac$ ✓

$\Delta = b^2 - 4ac = b^2 - 4b^2 = -3b^2$ ✓

$b > 0$, so $b^2 > 0$ and $-3b^2 < 0$. $\Delta < 0$, therefore x is non-real ✓

Equivalent: $\Delta = ac - 4ac = -3ac < 0$ because a and c are positive.

[149]

QUESTION 2

Quadratic Patterns

2.1.1 First differences: $x - 2$; $12 - x$; $x + 6$ ✓ (4)

Second differences: $-2x + 14$ and $2x - 6$ ✓

$-2x + 14 = 2x - 6$ ✓

$x = 5$ ✓

The pattern is 2 ; 5 ; 12 ; 23 ; ...

2.1.2 First differences: 3 ; 7 ; 11 ; ... so $a = 3$ and $d = 4$ (4)

$S_n = \frac{n}{2}[2(3) + (n-1)(4)] = 2n^2 + n$ ✓

$2n^2 + n > 500$ ✓

$n > 15{,}56$ ✓

$n = 16$ first differences, so it is the 17th term ✓

The sum of the first 16 first differences is 528. The sum of the first 15 is 465, which is too small.

2.2.1 80 ; 62 ✓✓ (2)

The first differences continue -34 ; -30 ; -26 ; -22 ; -18. One mark for each term.

2.2.2 First differences: -34 ; -30 ; -26 ✓ (2)

Second difference $= 4$ ✓

2.2.3 $2a = 4$, so $a = 2$ ✓ (4)

$3a + b = -34$, so $b = -40$ ✓

$a + b + c = 192$, so $c = 230$ ✓

$T_n = 2n^2 - 40n + 230$ ✓

Accepted alternative: $T_n = T_1 + (n-1)d_1 + \frac{(n-1)(n-2)}{2}d_2$ or solving three equations from T_1, T_2, T_3.

2.2.4 $T_{25} = 2(25)^2 - 40(25) + 230$ ✓ (2)
$= 480$ ✓
Answer only: full marks.

2.2.5 $2n^2 - 40n + 230 = 1830$ ✓ (3)
$2n^2 - 40n - 1600 = 0$, so $n^2 - 20n - 800 = 0$ and $(n - 40)(n + 20) = 0$ ✓
$n = 40$ or $n = -20$ (rejected, $n > 0$). The 40th term ✓
The negative root must be rejected for the last mark.

2.2.6 $2n^2 - 40n + 230 = 38$ ✓ (4)
$2n^2 - 40n + 192 = 0$, i.e. $n^2 - 20n + 96 = 0$ ✓
$(n - 8)(n - 12) = 0$ ✓
$n = 8$ or $n = 12$ ✓
Check: $T_8 = T_{12} = 38$. Both terms are asked for.

2.2.7 $T'_n = 4n - 40 = 0$ ✓ (2)
$n = 10$: the 10th term ✓
Alternative: $n = -\dfrac{b}{2a} = \dfrac{40}{4} = 10$ (1 mark for the substitution, 1 for the answer). The least value is $T_{10} = 30$.

2.3.1 First differences: 11 ; 19 ; 27 ; ... so $a = 11$ and $d = 8$ ✓ (3)
$S_n = \dfrac{n}{2}[2(11) + (n - 1)(8)]$ ✓
$= \dfrac{n}{2}[8n + 14] = 4n^2 + 7n$ ✓

2.3.2 $1 + S_n = 6681$ ✓ (4)
$4n^2 + 7n - 6680 = 0$ ✓
$(n - 40)(4n + 167) = 0$ ✓
$n = 40$ (the other root is negative). So 40 first differences ✓
Check route: $T_n = 6681$ gives $n = 41$, so 40 first differences are added to T_1.

2.4.1 First differences: 15 ; 13 ; 11 ; ... so $T_{n+1} - T_n = -2n + 17$ ✓ (4)
$-2n + 17 = -51$ ✓
$n = 34$ ✓
Between T_{34} and T_{35} ✓
Alternative: solve $T_n = -51$ in the sequence of first differences to get the same $n = 34$.

2.4.2 $n = -\dfrac{b}{2a} = -\dfrac{18}{2(-1)} = 9$ ✓ (3)
$T_9 = -19$ ✓
Add 20 ✓
Alternative: $T'_n = -2n + 18 = 0$ gives $n = 9$, then $T_9 = -19$.

2.5.1 $3a + b = 9$, with $a = 2$: $6 + b = 9$ ✓ (2)
$b = 3$ ✓
Alternative: $T_2 - T_1 = 9$, i.e. $(8 + 2b + 11) - (2 + b + 11) = 9$, so $b = 3$.

2.5.2 The pattern is 16 ; 25 ; 38 ; 55 ; ... First differences: 9 ; 13 ; 17 ; ... ✓ (3)
Common difference $= 4$ (this is the second difference) ✓
$T_p = 4p + 5$ ✓
Alternative: $T_p = 9 + (p - 1)(4) = 4p + 5$. Answer only: full marks.

2.5.3 (3)

$193 = 4p + 5$ ✓

$p = 47$ ✓

T_{47} and T_{48} ✓

Check: $T_{48} - T_{47} = 4763 - 4570 = 193$.

2.6.1 (2)

Second difference $= 15 - 7 = 8$, so the first differences are $7\ ;\ 15\ ;\ 23\ ;\ 31$ ✓

$T_5 = 4 + 7 + 15 + 23 + 31 = 80$ ✓

Alternative: find $a = 4$, $b = -5$, $c = 5$ and substitute $n = 5$ (2 marks).

2.6.2 (3)

$2a = 8$, so $a = 4$ ✓

$12 + b = 7$, so $b = -5$ ✓

$4 + (-5) + c = 4$, so $c = 5$ ✓

2.6.3 (3)

$T'_n = 8n - 5$ ✓

$8n - 5 > 0$, so $n > \dfrac{5}{8}$ ✓

So T_n is increasing for all $n \geq 1$ ✓

Alternative: the turning point is at $n = -\dfrac{b}{2a} = \dfrac{5}{8}$, which is less than 1; so for $n \in \mathbb{N}$ the least term is T_1 and the pattern increases.

2.7.1 (3)

First difference $= 4n - 2$ ✓

$T_{80} - T_{79} = 4(79) - 2 = 314$ ✓

$T_{79} = 12\,499 - 314 = 12\,185$ ✓

The 79th first difference is the difference between the 79th and 80th terms.

2.7.2 (5)

Second difference $= 4$ ✓

$2a = 4$, so $a = 2$ ✓

$3a + b = 2$, so $b = -4$ ✓

$9a + 3b + c = 25$ ✓

$c = 19$, so $T_n = 2n^2 - 4n + 19$ ✓

Alternative for c: $a + b + c = T_1 = 25 - 2 - 6 = 17$, so $c = 19$. Marks: second difference; a; b; method for c; c.

2.8.1 (2)

Heights: $28\ ;\ 54\ ;\ 76\ ;\ 94\ ;\ 108$. First differences: $26\ ;\ 22\ ;\ 18\ ;\ 14$ ✓

$T_5 = 108$ m ✓

2.8.2 (3)

$2a = -4$, so $a = -2$ ✓

$3(-2) + b = 26$, so $b = 32$ ✓

$-2 + 32 + c = 28$, so $c = -2$ ✓

Alternative for c: $c = T_1 - d_1 + D = 28 - 26 + (-4) = -2$, where d_1 is the first of the first differences and D is the second difference. The same task (show the general term) was asked in 2021 for a plain list of terms, with no context.

2.8.3 (3)

$n = -\dfrac{b}{2a} = -\dfrac{32}{2(-2)}$ ✓

$n = 8$ ✓

$T_8 = 126$ m ✓

Alternative: $T'_n = -4n + 32 = 0$ gives $n = 8$, then substitute.

2.8.4 (2)

The pattern is symmetrical about $n = 8$: $\dfrac{n + 3}{2} = 8$ ✓

$n = 13$ seconds ✓

Check: $-2n^2 + 32n - 2 = 76$ gives $(n - 3)(n - 13) = 0$.

QUESTION 3

3.1.1 $T_4 = 41$ ✓ (1)

Answer only: full marks.

3.1.2 $a = 62$ and $d = -7$ ✓ (3)

$-85 = 62 + (n-1)(-7)$ ✓

$n = 22$ ✓

Accepted alternative: $-7n + 69 = -85$, so $n = 22$.

3.1.3 Negative terms: -1 ; -8 ; ... ; -85, so $n = 13$ ✓ (3)

$a = -1$ ✓

$S_{13} = \frac{13}{2}[-1 + (-85)] = -559$ ✓

Accepted: $S_{22} - S_9 = -253 - 306 = -559$, or writing out all 13 negative terms and adding them.

3.1.4 Terms divisible by 5: 55 ; 20 ; -15 ; ... so $d = -35$ ✓ (4)

$T_m = 55 + (m-1)(-35) = -35m + 90$ ✓

The last such term is $-2017 + 7 = -2010$, so $-2010 = -35m + 90$ ✓

$m = 60$: 60 terms ✓

Alternative: $T_n = -7n + 69$ with $T_n = -2010$ gives $n = 297$ and the terms are T_2, T_7, ... so $\frac{297 - 2}{5} + 1 = 60$.

3.2.1 $-5 + 3d = 22$ ✓ (4)

$d = 9$ ✓

$x = -5 + 9 = 4$ ✓

$y = -5 + 2(9) = 13$ ✓

Alternative: two equations in x and y (for example $x - (-5) = y - x$ and $y - x = 22 - y$) solved simultaneously earn the same marks.

3.3.1 $(3k + 3) - (2k + 1) = (5k + 2) - (3k + 3)$ ✓ (5)

$k = 3$ ✓

The sequence is 7 ; 12 ; 17 ; ... so $a = 7$ ✓

$d = 5$ ✓

$T_{15} = 7 + 14(5) = 77$ ✓

3.3.2 The sequence 7 ; 12 ; 17 ; 22 ; ... : every second term is even, so the even terms are 12 ; 22 ; 32 ; ... (4)

$a = 12$ and $d = 10$ ✓✓

$S_{30} = \frac{30}{2}[2(12) + 29(10)]$ ✓

$= 4710$ ✓

Using $T_{30} = 12 + 29(10) = 302$ and $S_{30} = \frac{30}{2}(12 + 302)$ is also accepted.

3.4.1 $a = 9$ and $d = 4$ ✓ (3)

$T_{85} = 9 + (85 - 1)(4)$ ✓

$= 345$ ✓

Alternative: $T_n = 4n + 5$, so $T_{85} = 4(85) + 5 = 345$.

3.4.2 (2)

$S_{85} = \frac{85}{2}[2(9) + (85-1)(4)]$ ✓

$= 15\,045$ ✓

Alternative: $S_{85} = \frac{85}{2}(9 + 345) = 15\,045$.

3.4.3 (3)

$9 + (n-1)(4) = 525$ ✓

$4n + 5 = 525$ ✓

$n = 130$ ✓

3.5.1 (6)

$S_n = \frac{n}{2}[2(20) + (n-1)(-3)]$ ✓

$S_n = \frac{n}{2}(43 - 3n)$ ✓

$\frac{n}{2}(43 - 3n) = 2n^2 - 6n + 39$ ✓

$7n^2 - 55n + 78 = 0$ ✓

$(n-6)(7n-13) = 0$ ✓

$n = 6$ or $n = \frac{13}{7}$ (not a whole number, rejected). So $n = 6$ ✓

Check: $S_6 = 3(25) = 75$ and $T_6 = 75 = 75$.

3.6.1 (2)

$S_{25} = \frac{25}{2}[2(4) + (25-1)(3)]$ ✓

$= 1000$ ✓

3.6.2 (4)

$T_n = 4 + (n-1)(3) = 3n + 1$ ✓

$S_{80} - S_{25} = 9800 - 1000 = 8800$ ✓

$\sum_{n=26}^{80}(3n+1) = 8800$ ✓✓

Alternative: $T_n = 3n + 76$ for the added terms, so $\sum_{n=1}^{55}(3n + 76) = 8800$ (same 4 marks). Marks: general term; $S_{80} - S_{25}$; limits and expression in sigma notation; the right-hand side.

3.7.1 (1)

$5 + 9 + 13 + \ldots + 12\,001$ ✓

3.7.2 (4)

$T_1 = -1$ ✓

$T_{2999} = -4(2999) + 3 = -11\,993$ ✓

$(5 + 9 + \ldots + 11\,993 + 11\,997 + 12\,001) + (-1 - 5 - 9 - \ldots - 11\,993)$ ✓

$= 11\,997 + 12\,001 - 1 = 23\,997$ ✓

OR: $S_{3000} = 18\,009\,000$ ($a = 5$, $d = 4$) and $S_{2999} = -17\,985\,003$ ($a = -1$, $d = -4$); T_1 and T_{2999} (1 each, in place of the first two ticks); sum (1); answer (1).

3.8.1 (5)

$T_1 = \frac{1}{8}$ and $T_2 = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$, so $d = \frac{1}{4}$ ✓✓

$\frac{81}{8} = \frac{n}{2}\left[2\left(\frac{1}{8}\right) + (n-1)\left(\frac{1}{4}\right)\right]$ ✓

$81 = n^2$, so $n = 9$ ✓

$k = n - 1 = 8$ ✓

The sum starts at $p = 0$, so there are $k + 1$ terms. Alternative: $a = \frac{1}{8}$, $l = \frac{1}{4}k + \frac{1}{8}$ and $n = k + 1$, giving $\frac{81}{8} = \frac{(k+1)^2}{8}$ and $k = 8$.

3.9.1 (2)

$d = 3$, so $T_{17} - T_8 = 9 \times 3 = 27$ ✓✓

Alternative: $(3(17) + 2) - (3(8) + 2) = 53 - 26 = 27$ (1 mark for the substitution, 1 for the answer).

3.9.2 (5)

Number of terms $n = 91 - k$ ✓

Last term $T_{90} = 3(90) + 2 = 272$ ✓

$11\,102 = \frac{91-k}{2}[(3k+2)+272]$ ✓

$3k^2 + k - 2730 = 0$ ✓

$(k-30)(3k+91) = 0$, so $k = 30$ (since $k \neq -\frac{91}{3}$) ✓

Alternative: $S_{90} - S_{k-1} = 11\,102$ with $S_{90} = 12\,465$ and $S_{k-1} = 1363$ gives $k - 1 = 29$ terms, so $k = 30$.

3.10 (3)

$\left(1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{9}\right)$ ✓

$-\left(\frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{9} + \frac{1}{10}\right)$ ✓

$= 1 - \frac{1}{10} = \frac{9}{10}$ ✓

3.11.1 (6)

Area of the shaded part at each step: $\frac{3}{8}$; $\frac{3}{4}$; $\frac{9}{8}$; ... ; $\frac{15}{4}$ ✓

$a = \frac{3}{8}$ ✓

$d = \frac{3}{8}$ ✓

$S_{10} = \frac{10}{2}\left[2\left(\frac{3}{8}\right) + (10-1)\left(\frac{3}{8}\right)\right]$ ✓

$= \frac{165}{8}$ m² for one side ✓

Both sides: $2 \times \frac{165}{8} = \frac{165}{4} = 41{,}25$ m² ✓

The area of step k is $k \times \frac{1}{2} \times \frac{3}{4}$ (the shaded column under step k). Alternative: $a = \frac{3}{8}$, $(1 + 2 + \ldots + 10) = 55$, $\times 2$.

3.12.1 (2)

$\frac{96}{x} = \frac{72}{96}$ ✓

$x = 128$ ✓

OR: $r = \frac{72}{96} = 0{,}75$ and $x = 96 \div 0{,}75 = 128$ (2 marks).

3.12.2 (2)

$r = 0{,}75$ ✓

$S_n = \frac{128\,(1-(0{,}75)^n)}{1-0{,}75} = \frac{128\,(1-(0{,}75)^n)}{0{,}25} = 512\,(1-(0{,}75)^n)$ ✓

3.12.3 (2)

$S_\infty = \frac{a}{1-r} = \frac{128}{1-0{,}75}$ ✓

$= 512$ ✓

OR: $S_\infty = \lim_{n\to\infty}\left[512\,(1-(0{,}75)^n)\right] = 512$ (limit expression 1, answer 1).

3.13.1 (2)

$S_\infty = \frac{a}{1-\frac{1}{4}} = 12$ ✓

$a = 9$ ✓

3.13.2 (2)

$T_6 = 9\left(\frac{1}{4}\right)^5$ ✓

$= \frac{9}{1024}$ ✓

3.13.3 (4)

$9 + \frac{9}{4} + \frac{9}{16} + \ldots = 11{,}8125$, so $r = \frac{1}{4}$ ✓

$$S_n = \frac{9\left(1 - \left(\frac{1}{4}\right)^n\right)}{1 - \frac{1}{4}} = 11{,}8125$$ ✓

$\left(\frac{1}{4}\right)^n = \frac{1}{64}$ ✓

$n = 3$ ✓

3.13.4 (3)

$\sum_{k=1}^{20} 9(4)^{1-k} = 9 + \frac{9}{4} + \frac{9}{16} + \ldots$ ✓

$\sum_{k=1}^{20} 144(4)^{-k} = 36 + 9 + \frac{9}{4} + \ldots$ ✓

$= 4p$ ✓

Alternative: $144(4)^{-k} = 4 \times 36(4)^{-k} = 4 \times 9(4)^{1-k}$, so the sum is $4p$.

3.14.1 (3)

$r = \frac{1}{3}$ ✓

$$S_6 = \frac{4\left(1 - \left(\frac{1}{3}\right)^6\right)}{1 - \frac{1}{3}}$$ ✓

$= 5{,}9918\ldots = 5{,}99$ ✓

3.14.2 (4)

$4\left(\frac{1}{3}\right)^{n-1} > \frac{4}{6561}$ ✓

$\left(\frac{1}{3}\right)^{n-1} > \left(\frac{1}{3}\right)^8$ ✓

$n - 1 < 8$, so $n < 9$ ✓

$n = 8$ ✓

Alternative: write the terms as $4; \frac{4}{3}; \frac{4}{9}; \ldots$ and compare with $\frac{4}{6561}$ using logarithms. The inequality sign must be turned around where needed. Largest value: $n = 8$.

3.15.1 (2)

$T_6 = 5r^5 = 1215$ ✓

$r^5 = 243$, so $r = 3$ ✓

Answer only: full marks.

3.15.2 (4)

$16\,400 = \frac{5(3^n - 1)}{3 - 1}$ ✓

$3^n = 6561$ ✓

$n = 8$ ✓

2 more terms must be added ✓

Alternative: $S_6 = 1820$ and $16\,400 - 1820 = 14\,580$, then solve for the number of added terms. The answer is 2 terms.

3.15.3 (3)

$1215r^5 = 5$, so $r = \frac{1}{3}$ ✓

$S_\infty = \frac{a}{1-r} = \frac{1215}{1-\frac{1}{3}}$ ✓

$= 1822{,}5$ ✓

Answer only: full marks.

3.16.1 (3)

$n = 10$ and $r = 0{,}8$ ✓

$T_{10} = 24(0{,}8)^9$ ✓

$= 3{,}22$ cm ✓

3.16.2 (4)

$n = 18$ (each rectangle and its gap use 2 cm of the 36 cm) ✓

$S_{18} = \frac{24\left(1-(0{,}8)^{18}\right)}{1-0{,}8} = 117{,}84$ ✓

Area of the paper $= 36 \times 24 = 864$ ✓

Percentage $= \frac{117{,}84}{864} \times 100\% = 13{,}64\%$ ✓

The grey area equals the sum of the lengths, because each width is 1 cm.

3.17.1 (2)

Second circle: 5 cm ✓

Third circle: $2{,}5$ cm ✓

Alternative: $10\left(\frac{1}{2}\right)^2 = 2{,}5$ cm (2 marks).

3.17.2 (4)

Areas: 100π ; 25π ; $\frac{100}{16}\pi$; ... so $a = 100\pi$ ✓

$r = \frac{1}{4}$ ✓

$S_8 = \frac{100\pi\left(1-\left(\frac{1}{4}\right)^8\right)}{1-\frac{1}{4}}$ ✓

$= 418{,}87$ cm² ✓

The area of each circle is a quarter of the previous one because the radius is halved.

3.17.3 (4)

Radius $= \frac{1}{2}\left(\frac{5}{64}\right) = \frac{5}{128}$ ✓

$\frac{5}{128} = 10\left(\frac{1}{2}\right)^{n-1}$ ✓

$\left(\frac{1}{2}\right)^{n-1} = \frac{1}{256} = \left(\frac{1}{2}\right)^8$ ✓

$n - 1 = 8$, so $n = 9$ ✓

Alternative: use the diameter: $20\left(\frac{1}{2}\right)^{n-1} = \frac{5}{64}$ gives $n = 9$ (same 4 marks).

3.18.1 (2)

$a + ar = 5$ ✓

$a = \frac{5}{1+r}$ ✓

3.18.2 (6)

$S_\infty = T_1 + T_2 + \sum_{n=3}^{\infty} T_n = 5 + 4 = 9$ ✓

$\dfrac{a}{1-r} = 9$ ✓

$\dfrac{5}{1+r} \cdot \dfrac{1}{1-r} = 9$ ✓

$9r^2 = 4$, i.e. $r^2 = \dfrac{4}{9}$ ✓

$r = \dfrac{2}{3}$ ($r = -\dfrac{2}{3}$ rejected: positive terms) ✓

$a = 3$ ✓

Check: $3 + 2 = 5$ and $\sum_{n=3}^{\infty} T_n = \dfrac{3\left(\frac{2}{3}\right)^2}{1 - \frac{2}{3}} = 4$. Alternative: $r = \dfrac{9 - 4a}{9}$ substituted into $a + ar = 5$ gives a quadratic in a with the same solution.

3.19.1 (3)

$20 + 10 + 5 + \ldots$ ✓

$r = \dfrac{10}{20} = \dfrac{1}{2}$ ✓

$-1 < r < 1$, so the series converges ✓

3.19.2 (5)

$a = 5(2)^{3-p}$ and $r = \dfrac{1}{2}$ ✓

$\dfrac{5(2)^{3-p}}{1 - \frac{1}{2}} = \dfrac{5}{8}$ ✓

$5(2)^{4-p} = \dfrac{5}{8}$, so $(2)^{4-p} = \dfrac{1}{8}$ ✓

$(2)^{4-p} = (2)^{-3}$ ✓

$p = 7$ ✓

3.20.1 (1)

$T_n = 5(3)^{n-1}$ ✓

3.20.2 (4)

$\sum_{p=1}^{k} \dfrac{5}{3}(3)^p = 5 + 15 + 45 + \ldots$ ✓

$n = k$ ✓

$\dfrac{5\left(3^k - 1\right)}{3 - 1} = 49\,205$ ✓

$3^k = 19\,683 = 3^9$, so $k = 9$ ✓

3.20.3 (5)

$S_{20} = \dfrac{20}{2}[2a + 19(2)]$ ✓

$S_{20} = 20a + 380$ ✓

$S_\infty = \dfrac{a}{1 - \frac{1}{4}} = \dfrac{4a}{3}$ ✓

$20a + 380 = \dfrac{4a}{3} + 436$ ✓

$60a + 1140 = 4a + 1308$, so $56a = 168$ and $a = 3$ ✓

3.21.1 (3)

$\dfrac{t-4}{t+32} = \dfrac{t+8}{t-4}$ ✓

$(t-4)^2 = (t+8)(t+32)$ ✓

$t^2 - 8t + 16 = t^2 + 40t + 256$ ✓

$-48t = 240$, so $t = -5$

3.21.2 With $t = -5$: $a = 27$ and $T_2 = -9$ ✓ (3)

$r = \dfrac{-9}{27} = -\dfrac{1}{3}$ ✓

$T_{25} = 27\left(-\dfrac{1}{3}\right)^{24} = 3^3 \cdot 3^{-24} = 3^{-21}$ ✓

Accepted equivalent forms: $\left(\dfrac{1}{3}\right)^{21}$, $(9)^{-\frac{21}{2}}$, $(27)^{-7}$ and $(3^{21})^{-1}$.

3.21.3 $S_\infty = \dfrac{a}{1-r} = \dfrac{27}{1-\left(-\dfrac{1}{3}\right)}$ ✓ (2)

$= \dfrac{81}{4} = 20{,}25$ ✓

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QUESTION 4

Functions: Hyperbola

4.1 $x = 3$ ✓ (2)

$y = 1$ ✓

4.2 $x \in \mathbb{R}, x \neq 3$ ✓ (1)

Also accepted: $x < 3$ or $x > 3$, or $(-\infty; 3) \cup (3; \infty)$.

4.3 $y \in \mathbb{R}, y \neq 1$ ✓ (1)

Also accepted: $y < 1$ or $y > 1$, or $(-\infty; 1) \cup (1; \infty)$.

4.4 $y = 0$: $0 = \dfrac{-2}{x-3} + 1$ ✓ (2)

$\dfrac{-2}{x-3} = -1$ so $x - 3 = 2$

$x = 5 \quad (5; 0)$ ✓

4.5 $x = 0$: $f(0) = \dfrac{-2}{0-3} + 1$ ✓ (2)

$= \dfrac{5}{3} \quad (0; \dfrac{5}{3})$ ✓

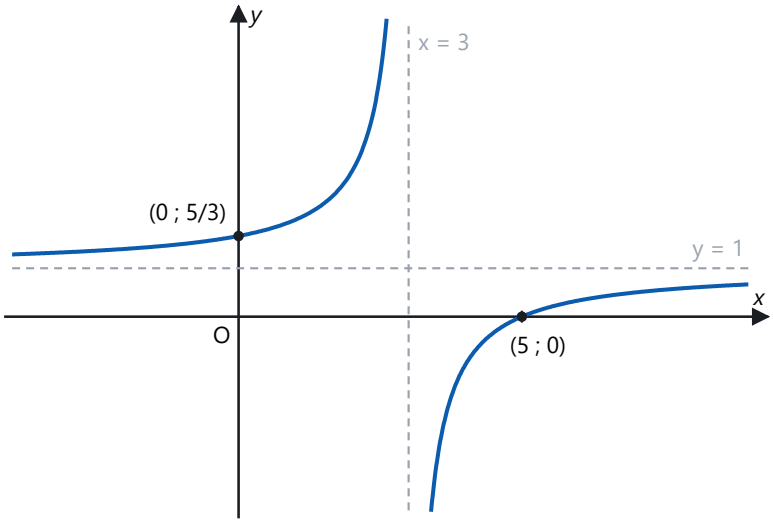
4.6 (3)

Both asymptotes drawn as dotted lines, $x = 3$ and $y = 1$ ✓

Intercepts $(5; 0)$ and $(0; \frac{5}{3})$ marked ✓

Shape: one branch above $y = 1$ to the left of $x = 3$, the other below $y = 1$ to the right of $x = 3$ ✓

Because $a < 0$ the branches lie in the 'top-left' and 'bottom-right' corners formed by the asymptotes.



4.7 (2)

$f'(x) = \frac{2}{(x-3)^2} > 0$ for all $x \neq 3$

So $x \geq 0$ ✓

and $x \neq 3$ ✓

Answer: $0 \leq x < 3$ or $x > 3$

A table or sign diagram of x and $f'(x)$ is also accepted.

4.8 (3)

$\frac{-2}{x-3} + 1 \geq 0$, i.e. $f(x) \geq 0$ ✓

$x < 3$ ✓

or $x \geq 5$ ✓

Reasoning: for $x < 3$ the graph lies above $y = 1$, so $f(x) > 0$. For $x > 3$ the graph is at or above the x-axis only from the x-intercept $x = 5$ onwards.

> **Memo trap:** Multiplying both sides by $x - 3$ without turning the inequality sign for $x < 3$.

4.9 (1)

$p = -2$ ✓

The vertical asymptote is $x = 2$, so $x + p = x - 2$.

4.10 (2)

The asymptotes meet on the axis of symmetry, at $x = 2$ ✓

$y = 2 + 1 = 3$: the horizontal asymptote is $y = 3$ ✓

Answer only: 2 marks.

4.11 (2)

Substitute $(0; 2)$: $2 = \frac{a}{0-2} + 3$ ✓

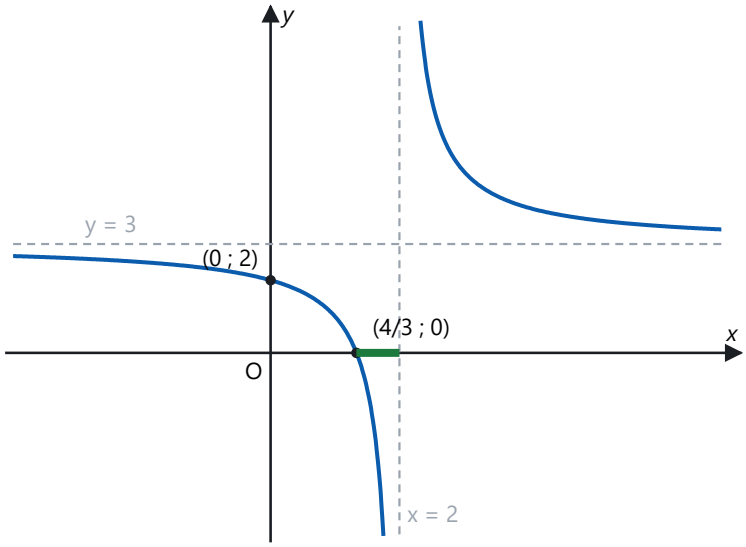
$-1 = \frac{a}{-2}$ so $a = 2$ ✓

4.12 (3)

x-intercept: $0 = \dfrac{2}{x-2} + 3$ so $x = \dfrac{4}{3}$ ✓

$\dfrac{4}{3} \leq x < 2$ ✓✓

Any correct notation: $x \in [\frac{4}{3}; 2)$. The strict inequality at $x = 2$ is needed for the second mark.



4.13 Reflection about the line $x = 2$ ✓✓ (2)

Any ONE of these, 2 marks each: reflection about the line $y = 3$; reflection in the y-axis followed by a translation of 4 units to the right (1 + 1); reflection in the x-axis followed by a translation of 6 units up (1 + 1).

Why: $f'(x) = \dfrac{-2}{(x-2)^2}$ is negative. Reflecting about $x = 2$ gives $h(x) = \dfrac{-2}{x-2} + 3$, which has $h'(x) > 0$.

4.14 $(-p; 0)$ ✓ (1)

The vertical asymptote is $x = -p$, and it passes through the x-intercept of g.

4.15 (5)

$g(-p) = 0$, so $-p + c = 0$ and $p = c$ ✓

$g(3) = 3 + c$, so $q = 3 + c$ ✓

$f(x) = \dfrac{a}{x+c} + 3 + c$. At the y-axis $f(0) = g(0)$: $\dfrac{a}{c} + 3 + c = c$, so $a = -3c$ ✓

At $x = 5$: $\dfrac{-3c}{5+c} + 3 + c = 5 + c$, so $-3c = 10 + 2c$ ✓

$c = -2$, $a = 6$, $p = -2$, $q = 1$: $f(x) = \dfrac{6}{x-2} + 1$ ✓

Alternative: the point $(5; 5 + c)$ is substituted into $f(x) = \dfrac{a}{x+c} + 3 + c$ together with $a = -3c$; marks are the same.

> **Memo trap:** Students forget that $p = c$ and leave three unknowns. The condition 'meet on the x-axis' is what gives $p = c$.

4.16 The asymptotes of f meet at $(2; 1)$. $g(x) = x - 2$ passes through $(2; 0)$, so g must move up 1 unit to give $y = x - 1$ ✓ (2)

Translation of 1 unit up ✓

Also accepted: translation of 1 unit to the left, because $g(x + 1) = x - 1$.

The other axis of symmetry, $y = -x + 3$, has a negative gradient, so it cannot come from g by a translation.

4.17 $p = -2$ ✓ (2)

$q = -1$ ✓

The vertical asymptote is $x = -p = 2$ and the horizontal asymptote is $y = q = -1$.

4.18 $m = -1$ and the axis passes through $(2; -1)$: $-1 = -(2) + c$ ✓ (2)

$y = -x + 1$ ✓

The other axis of symmetry has a positive gradient: $y = x + t$ with $-1 = 2 + t$, so $t = -3$ and $y = x - 3$. If the question gives $y = x + t$ and asks for t, substitute $(2; -1)$ (1) and find $t = -3$ (1).

Answer only: 2 marks.

4.19 $x = 2 - 4 = -2$ ✓ (1)

Alternative: $k(x) = \dfrac{1}{x + 4 - 2} - 1 = \dfrac{1}{x + 2} - 1$, so the asymptote is $x = -2$.

4.20 Translation 2 units to the left: $x = 3 - 2$ ✓ (2)

$x = 1$ ✓

Alternative: $g(x) = \dfrac{1}{x} - 1 = 0$ gives $x = 1$. Answer only: 2 marks.

4.21 Critical values $x = -4$ and $x = -1$ ✓ (2)

$x \leq -4$ or $x > -1$ ✓

In interval notation: $(-\infty; -4] \cup (-1; \infty)$.

> **Memo trap:** Writing $-4 \leq x < -1$. That is where the graph is BELOW the x-axis.

4.22 $y + t = \dfrac{6}{x - p}$, so $y = \dfrac{6}{x - p} - t$ ✓ (4)

The asymptotes are $x = p$ and $y = -t$ ✓

The asymptotes meet at $B(-1; 2)$: $p = -1$ ✓

$-t = 2$, so $t = -2$ ✓

Check with A: $(-4 + 1)(0 - 2) = 6$.

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QUESTION 5

Functions: Exponential, Logarithmic And Inverse

5.1 $R(0; 1)$ ✓ (1)

$a^0 = 1$ for every $a > 0$.

5.2 $25 = a^2$ ✓ (2)

$a = 5$ (since $a > 0$) ✓

5.3 $\dfrac{1}{125} = 5^b$ ✓ (4)

$5^{-3} = 5^b$ ✓

$b = -3$ ✓

$DP = 2 - (-3) = 5$ units ✓

5.4 $h(x + 1) = -k$. For $x < -4$: $x + 1 < -3$ ✓ (3)

so $0 < 5^{x+1} < \dfrac{1}{125}$, i.e. $0 < -k < \dfrac{1}{125}$ ✓

$-\dfrac{1}{125} < k < 0$ ✓

h is always positive, so $-k$ must be positive: that gives the upper end $k < 0$.

5.5 (3)

$16 = 2^{-x}$ ✓
$2^4 = 2^{-x}$ ✓
$x = -4 \quad B(-4;16)$ ✓

5.6 (2)

$x \in (0;\infty)$ or $x > 0$ ✓✓

The domain of f^{-1} is the range of f. Since $2^{-x} > 0$, the range of f is $y > 0$.

5.7 (3)

$g(x) = 32 \cdot 2^{-x} = 2^5 \cdot 2^{-x}$ ✓
$= 2^{-(x-5)}$ ✓
5 units to the right ✓

Alternative: $f(0) = 1$ moves to $g(5) = 1$, so the graph moves 5 units right.

5.8 (3)

$\dfrac{32}{2^x} < 1$ so $2^x > 32$ ✓
$2^x > 2^5$ ✓
$x > 5$ ✓

Alternative: $g(5) = 1$ and g is decreasing, so $g(x) < 1$ for $x > 5$.

5.9 (1)

$y = -9$ ✓

For $h(x) = a^x$ the asymptote is $y = 0$.

5.10 (2)

$y = 0$: $0 = 3^x - 9$ so $3^x = 3^2$ ✓
$x = 2 \quad B(2;0)$ ✓

5.11 (3)

$f(0) = 3^0 - 9 = -8$, so $A(0;-8)$ ✓
$m = \dfrac{0-(-8)}{2-0} = 4$ ✓
$k(x) = 4x - 8$ ✓

5.12 (3)

$k(1) = 4(1) - 8 = -4$ ✓
$f(1) = 3^1 - 9 = -6$ ✓
distance $= -4 - (-6) = 2$ units ✓

5.13 (1)

$g(x) = 3^x, \quad x \in [-1;3)$ ✓

5.14 (2)

Range of g: $y \in \left[\dfrac{1}{3};27\right)$ ✓
Domain of g^{-1}: $x \in \left[\dfrac{1}{3};27\right)$ ✓

$g(-1) = \dfrac{1}{3}$ (closed end) and $g(3) = 27$ (open end).

5.15 (2)

$x = 3^y$ ✓
$y = \log_3 x, \quad x \in \left[\dfrac{1}{3};27\right)$ ✓

5.16 (2)

$-\dfrac{15}{4} = a^2 - 4$ ✓
$a^2 = \dfrac{1}{4}$ so $a = \dfrac{1}{2}$ (since $a > 0$) ✓

5.17 (1)

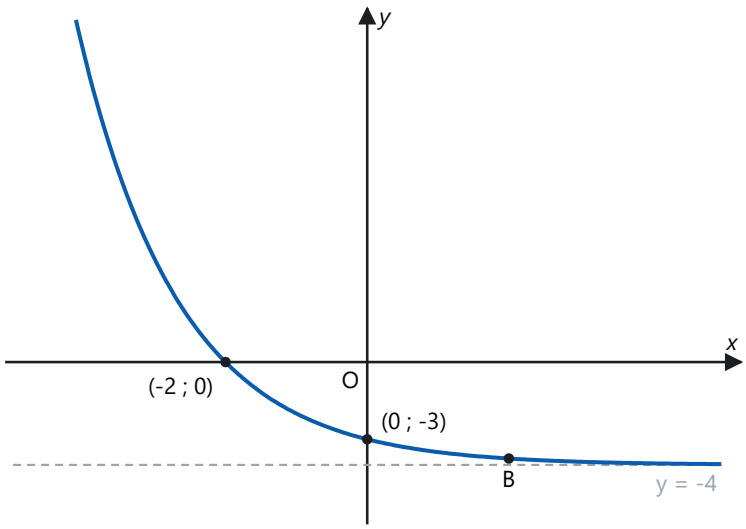
$y > -4$ or $y \in (-4;\infty)$ ✓

5.18 Intercepts $(-2;0)$ and $(0;-3)$ ✓ (3)

Asymptote $y = -4$ (dotted) ✓

Decreasing curve through $B(2;-3{,}75)$ ✓

x-intercept: $0 = (\frac{1}{2})^x - 4$ gives $(\frac{1}{2})^x = 4$ and $x = -2$.



5.19 $4 = (\frac{1}{2})^x - 4$ so $(\frac{1}{2})^x = 8$ ✓ (3)

$x = -3 \quad C(-3;4)$ ✓

$C'(4;-3)$ ✓

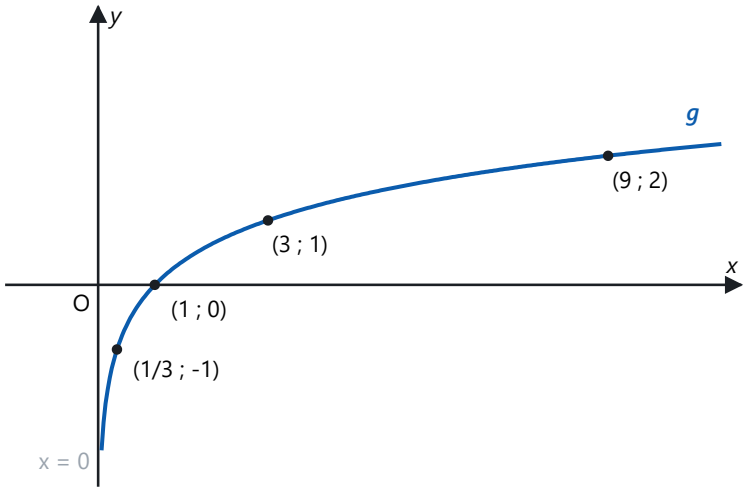
5.20 $f: \ y = 3^x$. Reflect: $x = 3^y$ ✓ (2)

$y = \log_3 x$ ✓

5.21 Vertical asymptote $x = 0$ (the y-axis) ✓ (4)

Increasing curve ✓

Any two correct points, e.g. $(1;0)$, $(3;1)$, $(9;2)$, $(27;3)$ ✓✓



5.22 $f(x) > 0$ for all x, so $g(x) > 0$ is needed: $x > 1$ ✓ (2)

$x \in (1;\infty)$ ✓

5.23 (2)

$\log_3 x = -1$ gives $x = \frac{1}{3}$ ✓

$0 < x \leq \frac{1}{3}$ ✓

The domain of g is $x > 0$, so the lower end $x = 0$ is NOT included.

5.24 (4)

$3^x - 3^{-x} = \frac{8}{3}$ ✓

Let $t = 3^x$: $t - \frac{1}{t} = \frac{8}{3}$ so $3t^2 - 8t - 3 = 0$ ✓

$(3t + 1)(t - 3) = 0$ ✓

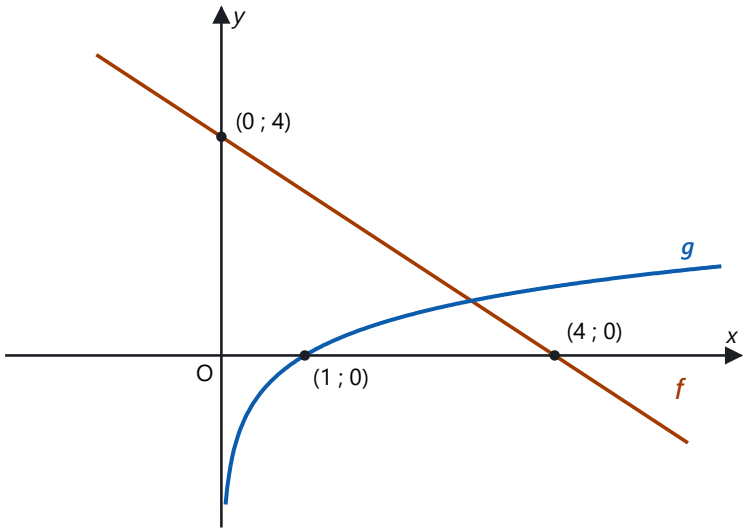
$3^x = -\frac{1}{3}$ (no solution) or $3^x = 3$, so $x = 1$ ✓

> **Memo trap:** Writing 3^{-x} as -3^x or $\frac{1}{3^x}$ incorrectly. Multiply every term by 3^x to remove the fraction.

5.25 (4)

f: straight line with negative gradient: $(4; 0)$ ✓ and $(0; 4)$ ✓

g: increasing curve with the y-axis as asymptote ✓, through $(1; 0)$ ✓



5.26 (2)

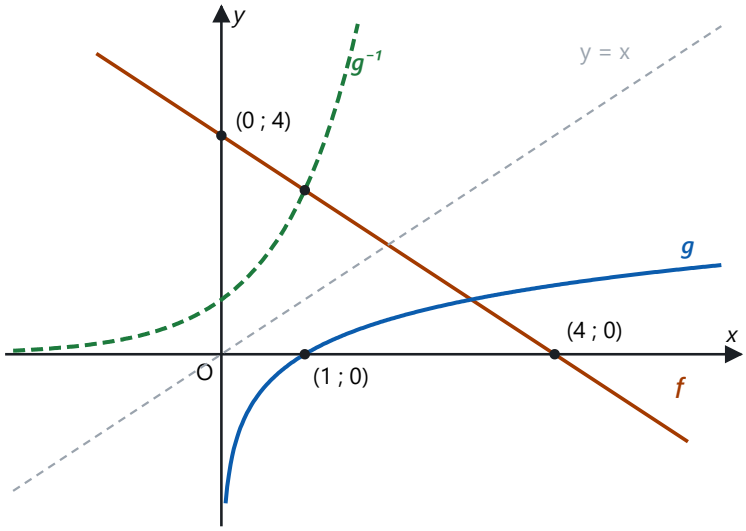
$x = \log_3 y$ ✓

$y = 3^x$ ✓

5.27 $\log_3(4-x) = x$ means $4 - x = 3^x$ ✓✓ (3)

The solution is the x-coordinate of the point of intersection of f and g^{-1} ✓

Reflect g about the line $y = x$ to get g^{-1}, then read off where it crosses f.



5.28 $x = 1$ ✓ (1)

Check: $\log_3 3 = 1$.

5.29 $3 = \log_2 k$ ✓ (2)

$k = 2^3 = 8$ ✓

5.30 $-2 = \log_2 x$ gives $x = \frac{1}{4}$ ✓ (2)

$\frac{1}{4} \leq x \leq 8$ ✓

5.31 $f^{-1}(x) = 2^x > 0$ for all x ✓ (2)

so $x < 0$ ✓

$x \in (-\infty; 0)$ is also accepted.

5.32 $t = \log_{1/2} 4 = -2$ ✓ (1)

$(\frac{1}{2})^{-2} = 4$.

5.33 $A(1; 0)$ ✓ (1)

5.34 $x = \log_{1/2} y$ ✓ (2)

$y = (\frac{1}{2})^x$ ✓

$y = 2^{-x}$ is also accepted.

5.35 $y = 0$ ✓ (1)

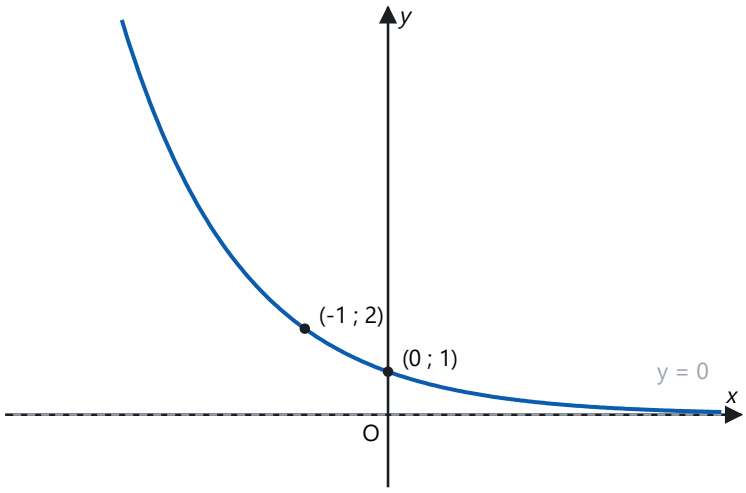
5.36 (3)

Decreasing exponential curve with asymptote $y = 0$ (the x-axis) ✓

y-intercept $(0; 1)$ ✓

Any other correct point, e.g. $(-1; 2)$ or $(-2; 4)$ or $(1; \frac{1}{2})$ ✓

The graph has no x-intercept.



5.37 (2)

$q(x) = (\frac{1}{2})^{x-2}$, so $q(3) = \frac{1}{2}$ ✓

$0 < y < \frac{1}{2}$ ✓

q is decreasing and never reaches the asymptote $y = 0$.

5.38 (3)

$q = -6$ ✓

$-15 = a(3)^2 - 6$ ✓

$-9 = 9a$, so $a = -1$ ✓

Hence $g(x) = -3^x - 6$.

5.39 (1)

$y < -6$ or $y \in (-\infty; -6)$ ✓

a is negative, so the graph lies BELOW its asymptote.

5.40 (2)

$y \in \mathbb{R}, y \neq -2$ ✓✓

Also accepted: $y < -2$ or $y > -2$.

5.41 (2)

$E(3; 0)$ and $D(3; -2)$, so $p = -3$ ✓

$q = -2$: $g(x) = \frac{6}{x - 3} - 2$ ✓

5.42 (3)

$f(t) = 1$ ✓

$\log_5 t = 1$ ✓

$t = 5$ ✓

Alternative: substitute A into g: $1 = \frac{6}{t - 3} - 2$, so $t - 3 = 2$ and $t = 5$.

5.43 (2)

$5^x < 5^2$ ✓

$x < 2$ ✓

$x \in (-\infty; 2)$ is also accepted.

5.44 Gradient $= -1$ and passes through $D(3; -2)$ ✓ (3)

$y = -(x - 3) - 2 = -x + 1$ ✓

Intersects the x-axis at $(1; 0)$, which is B on f: $B(1; 0)$ ✓

The axis with positive gradient is $y = x - 5$.

[100]

QUESTION 6

Functions: Parabola, Straight Line And Mixed Graphs

6.1.1 $f'(x) = -2ax + b$ ✓ (5)

$f'(-1) = 2a + b = 6$ ... (1) ✓

$f(-1) = -a - b + 12 = 7$, so $a + b = 5$ ... (2) ✓

(1) − (2): $a = 1$ ✓

Then $b = 5 - 1 = 4$ ✓

Alternative: $b = 6 - 2a$ substituted into $-a - b + 12 = 7$ gives $a = 1$ and $b = 4$ (same marks).

6.1.2 $-x^2 + 4x + 12 = 0$ ✓ (3)

$x^2 - 4x - 12 = 0$

$(x - 6)(x + 2) = 0$

$x = -2$ or $x = 6$

$(-2\,;\,0)$ ✓ and $(6\,;\,0)$ ✓

6.1.3 $f'(x) = -2x + 4 = 0$ ✓ (3)

$x = 2$ ✓

$y = -(2)^2 + 4(2) + 12 = 16$ ✓

Turning point: $(2\,;\,16)$

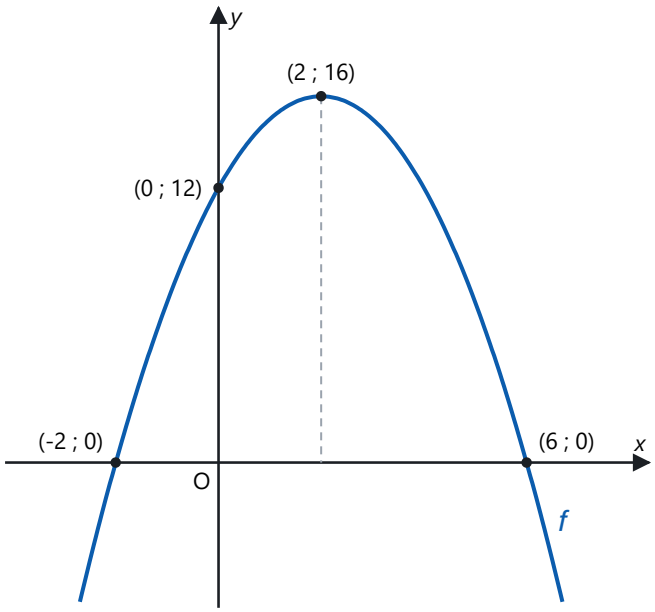
Also marked: $x = -\dfrac{b}{2a} = 2$, or completing the square $f(x) = -(x - 2)^2 + 16$, or the midpoint of the x-intercepts.

6.1.4 Downward-opening parabola (shape) ✓ (4)

x-intercepts $(-2\,;\,0)$ and $(6\,;\,0)$ ✓

y-intercept $(0\,;\,12)$ ✓

Turning point $(2\,;\,16)$ ✓



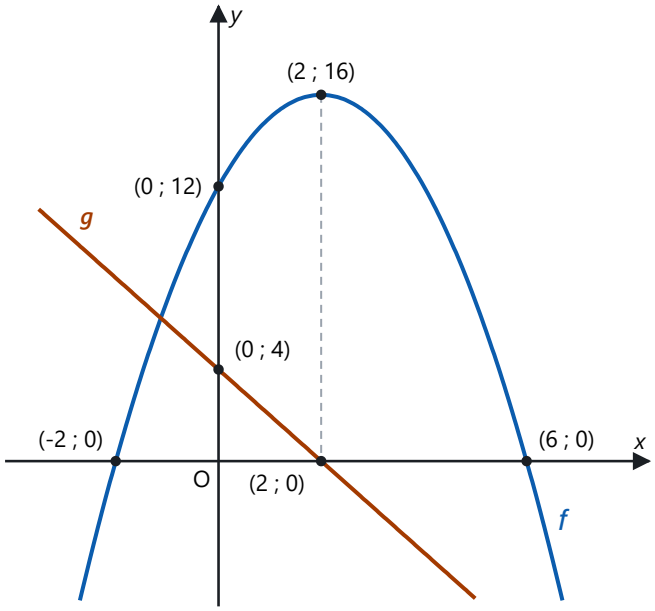
6.1.5 $f(x) = 12$ at $x = 0$ and (by symmetry about $x = 2$) at $x = 4$ ✓ (3)

$0 < x < 4$ ✓✓

The answer $x \in (0\,;\,4)$ is accepted.

6.1.6 x-intercept $(2\,;\,0)$ ✓ (2)

y-intercept $(0\,;\,4)$, straight line sloping downwards ✓



6.1.7 $f \geq 0$ for $-2 \leq x \leq 6$ and $g \geq 0$ for $x \leq 2$ (3)

$x \leq -2$ ✓

or $2 \leq x \leq 6$ ✓✓

Accepted: $x \in (-\infty\,;\,-2] \cup [2\,;\,6]$. Product ≤ 0 means one graph is above and the other below the x-axis, or at least one of them is zero.

> **Memo trap:** Leaving out the endpoints (the memo uses $\leq$ because the product may equal 0).

6.2.1 $f'(x) = -2x + 8 = 0$, so $x = 4$ ✓ (2)

$f(4) = -(4)^2 + 8(4) - 12 = 4$ ✓

Alternative: the x-intercepts are 2 and 6, so $x = \dfrac{2+6}{2} = 4$ (1), $y = 4$ (1).

6.2.2 $y \leq 4$, i.e. $(-\infty\,;\,4]$ ✓ (1)

6.2.3 Critical values $x = 1$ and $x = 4$ ✓ (2)

$x \leq 1$ or $x > 4$ ✓

$x = 4$ is NOT included: it is the asymptote.

6.2.4 The vertical asymptote of g is $x = 4$; shifting 3 units to the left gives $x = 4 - 3 = 1$ ✓ (1)

Also accepted: $h(x) = \dfrac{12}{x-1} + 4$, so $x = 1$.

6.2.5 $y = \dfrac{12}{x-p} - t$ ✓ (4)

Vertical asymptote $x = p$ and horizontal asymptote $y = -t$ ✓

The asymptotes meet at B$(4\,;\,4)$, so $p = 4$ ✓

$-t = 4$, so $t = -4$ ✓

Check with A: $(1-4)(0-4) = 12$.

6.2.6 x-intercepts of f: $x = 2$ and $x = 6$ ✓ (4)

$g'(x) < 0$ for all $x \neq 4$ (the branches of g are decreasing), so $f(x) \leq 0$ is needed

$x \leq 2$ ✓ or ✓ $x \geq 6$ ✓

Accepted: $(-\infty\,;\,2] \cup [6\,;\,\infty)$.

6.3.1 Yes ✓: every x-value has only one y-value (one-to-one mapping / vertical line test). ✓ (2)

6.3.2 $\mathrm{R}(8\,;\,-4)$ ✓ (1)

6.3.3 $8 = a(-4)^2$ ✓ (2)

$a = \dfrac{1}{2}$ ✓

6.3.4 $f: y = \dfrac{1}{2}x^2$; swap x and y: $x = \dfrac{1}{2}y^2$ ✓ (3)

$y^2 = 2x$ ✓

$y = -\sqrt{2x}$ ✓

The negative root is the only one that applies, because f^{-1} lies below the x-axis ($y \leq 0$).

6.4.1 $m = \dfrac{16-2}{7-0} = 2$ ✓ (2)

The y-intercept is $(0\,;\,2)$, so $c = 2$ and $g(x) = 2x + 2$ ✓

Alternative: $16 = m(7) + 2$ gives $m = 2$ (1); $c = 2$ (1).

6.4.2 $x = \dfrac{-b}{2a} = \dfrac{4}{2} = 2$ ✓ (3)

$f(2) = (2)^2 - 4(2) - 5 = -9$ ✓

$y \geq -9$, i.e. $[-9\,;\,\infty)$ ✓

Alternative: $y = (x-2)^2 - 9$ from completing the square.

6.4.3 $MN = (x^2 - 4x - 5) - (2x + 2) = x^2 - 6x - 7$ ✓ (4)

$x^2 - 6x - 7 = 9$ ✓

$x^2 - 6x - 16 = 0$; $(x-8)(x+2) = 0$; $x = 8$ or $x = -2$ ✓

T lies on the negative x-axis, so $x = -2$ and OT $= 2$ units ✓

$x = 8$ is rejected.

6.4.4 $y = 2(-2) + 2$ ✓ (2)

$\mathrm{N}(-2\,;\,-2)$ ✓

6.4.5 $f'(x) = 2x - 4$ ✓ (5)

$2x - 4 = 2$ ✓

$x = 3$ ✓

$f(3) = -8$ ✓

$y + 8 = 2(x-3)$, so $y = 2x - 14$ ✓

Alternative: $x^2 - 4x - 5 = 2x + p$, i.e. $x^2 - 6x - 5 - p = 0$; $\Delta = 0$ gives $36 + 4(5+p) = 0$, so $p = -14$ and $y = 2x - 14$ (same 5 marks).

6.4.6 $k < -14$ ✓ (1)

Lines with the gradient of the tangent that lie below the tangent miss the parabola.

6.5.1 $p = -2$ ✓ (1)

6.5.2 $\mathrm{D}(0\,;\,-8)$ (the y-intercept of f) ✓ (3)

Substitute $(0\,;\,-8)$ into g: $-8 = \dfrac{a}{0-2}$, so $a = 16$ ✓

Substitute $(2\,;\,0)$ into f: $0 = (2)^2 + b(2) - 8$, so $b = 2$ ✓

6.5.3 (3)

$m = \tan 45^\circ = 1$ ✓

$-9 = (1)(-1) + c$ ✓

$c = -8$, so $y = x - 8$ ✓

6.5.4 No ✓. The line passes through both C and D, so it cuts f twice. ✓ (2)

Alternative: a tangent at the turning point C has gradient 0, not 1.

6.5.5 (4)

$f(m - x) = f[-(x - m)]$: f is reflected in the y-axis, then moved m units to the right

The turning point $(-1\,;\,-9)$ moves to $(1 + m\,;\,-9)$ and then to $(1 + m\,;\,-9 + q)$

One x-intercept at $x = 0$ means the turning point is the point $(0\,;\,0)$

$1 + m = 0$, so $m = -1$ ✓✓

$-9 + q = 0$, so $q = 9$ ✓✓

Alternative: $h(x) = (m - x)^2 + 2(m - x) - 8 + q$; $h(0) = 0$ and the discriminant equal to 0 give $m = -1$ and $q = 9$ (2 + 2).

> **Memo trap:** Writing $m = 1$: $f(m - x)$ moves the graph to the RIGHT by m after the reflection, so the turning point x-value is $1 + m$.

6.6.1 $(-3\,;\,-8)$ ✓✓ (2)

6.6.2 (3)

$m = -3$ (the axis of symmetry is $x = -3$) ✓

$n = g(-3) = -\frac{1}{2}(-3) - \frac{7}{2}$ ✓

$n = -2$ ✓

6.6.3 (3)

OCDE is a trapezium with parallel sides OC $= \frac{7}{2}$ and ED $= 2$, and height OE $= 3$

Area $= \frac{1}{2}(DE + OC) \times OE$ ✓

$= \frac{1}{2}\left(2 + \frac{7}{2}\right)(3)$ ✓

$= \frac{33}{4} = 8{,}25$ square units ✓

Alternative: area of the rectangle (OE × ED) plus the triangle on top.

6.6.4 (5)

g^{-1}: $x = -\frac{1}{2}y - \frac{7}{2}$, so $g^{-1}(x) = -2x - 7$ ✓

$h(x) = -2x - 7 + k$ has gradient -2

$f'(x) = x + 3$ ✓

$x + 3 = -2$ ✓

$x = -5$ ✓

$y = \frac{1}{2}(-5 + 3)^2 - 8 = -6$; point of contact $(-5\,;\,-6)$ ✓

Alternative: equate $f(x) = h(x)$ and use $\Delta = 0$ to get $x = -5$ and $y = -6$.

6.7.1 B$(-5\,;\,0)$ ✓ and D$(7\,;\,0)$ ✓ (2)

6.7.2 $m_{\text{AE}} = \tan 14{,}04^\circ = 0{,}25$ ✓ (1)

6.7.3 (5)

$m_{\text{tangent}} \times m_{\text{AE}} = -1$, so $m_{\text{tangent}} = -4$ ✓

$f'(x) = 2x - 2$ (from $f(x) = x^2 - 2x - 35$) ✓

$2x - 2 = -4$ ✓

$x_T = -1$ ✓

$y_T = (-1)^2 - 2(-1) - 35 = -32$; T$(-1\,;\,-32)$ ✓

6.7.4 (6)

$m_g = \frac{1}{4}$ ✓

$y + 27 = \frac{1}{4}(x + 2)$ ✓

$y = \frac{1}{4}x - \frac{53}{2}$ ✓

$x^2 - 2x - 35 = \frac{1}{4}x - \frac{53}{2}$ ✓

$4x^2 - 9x - 34 = 0$ i.e. $(x + 2)(4x - 17) = 0$ ✓

$x = \frac{17}{4} = 4{,}25$ ($x \neq -2$, that is K) ✓

6.8.1 (1)

$y = -10$ ✓

$f(0) = -10$.

6.8.2 (2)

$x = -\frac{-8}{2(2)} = 2$ ✓

$f(2) = 2(2)^2 - 8(2) - 10 = -18$; D$(2\,;\,-18)$ ✓

Alternative: $f'(x) = 4x - 8 = 0$ gives $x = 2$, then $y = -18$.

6.8.3 (3)

$q = -10$ ✓

Substitute $(2\,;\,-18)$: $-18 = a(2)^2 - 10$ ✓

$a = -2$ ✓

6.8.4 (1)

$y < -10$, i.e. $y \in (-\infty\,;\,-10)$ ✓

The graph lies below its horizontal asymptote $y = -10$ (because $a < 0$).

6.8.5 (2)

The minimum value of f is -18 ✓

$f(x) - k > 0$ for all x when $k < -18$ ✓

6.9.1 (1)

$y = -4$ ✓

6.9.2 (2)

$x = 2y - 4$ (swap x and y) ✓

$g^{-1}(x) = \frac{1}{2}x + 2$ ✓

Answer only: full marks.

6.9.3 (3)

$\frac{1}{2}x + 2 = 2x - 4$ ✓

$x + 4 = 4x - 8$; $3x = 12$; $x = 4$ ✓

$y = 2(4) - 4 = 4$; A$(4\,;\,4)$ ✓

Alternative: A lies on $y = x$, so $2x - 4 = x$ gives $x = 4$ and $y = 4$ (same marks).

6.9.4 (2)

AB $= \sqrt{(4 - 0)^2 + (4 + 4)^2}$ ✓

$= \sqrt{80} = 4\sqrt{5} = 8{,}94$ units ✓

Answer only: full marks.

6.9.5 (5)

BC $= \sqrt{(0+4)^2+(-4-0)^2} = \sqrt{32} = 4\sqrt{2}$ ✓

AB $=$ AC $= \sqrt{80}$ (g and g^{-1} are reflections in $y = x$), so A lies on the perpendicular bisector of BC; the midpoint of BC is $(-2\,;\,-2)$ ✓

Height: $h = \sqrt{(4+2)^2+(4+2)^2} = \sqrt{72} = 6\sqrt{2}$ ✓

Area $= \frac{1}{2} \times BC \times h$ ✓

$= \frac{1}{2} \times 4\sqrt{2} \times 6\sqrt{2} = 24$ square units ✓

Alternative: D$(2\,;\,0)$ lies on AB. Area $= \Delta$BDC $+\Delta$ADC $= \frac{1}{2}(6)(4) + \frac{1}{2}(6)(4) = 24$ (1 mark for DC $= 6$, 1 each for the two areas, 1 for the substitution, 1 for the answer).

6.10.1 (2)

$f(0) = -\frac{1}{2}(0-2)^2 + 8$ ✓

$= 6$; C$(0\,;\,6)$ ✓

6.10.2 (1)

$8 = \frac{d}{2}$, so $d = 16$ ✓

Substitute B$(2\,;\,8)$.

6.10.3 (3)

$f \geq 0$ for $-2 \leq x \leq 6$; $g > 0$ for $x > 0$ and $g < 0$ for $x < 0$

$-2 \leq x < 0$ ✓✓

or $x \geq 6$ ✓

Accepted: $x \in [-2\,;\,0) \cup [6\,;\,\infty)$.

> **Memo trap:** Including $x = 0$: g is not defined there.

6.10.4 (5)

$-x + k = \frac{16}{x}$ ✓

$-x^2 + kx - 16 = 0$ ✓

$\Delta = k^2 - 4(-1)(-16)$ ✓

$k^2 - 64 < 0$ (no real roots) ✓

$-8 < k < 8$ ✓

Alternative (tangents): $g'(x) = -\frac{16}{x^2} = -1$ gives $x = \pm 4$, so the tangents are $y = -x + 8$ and $y = -x - 8$. $k = \pm 8$ and then the inequality (same 5 marks).

> **Memo trap:** Writing $k < -8$ or $k > 8$: that is where the line DOES cut the hyperbola.

6.10.5 (4)

The tangent has $k = 8$, so $h(x) = -x + 8$

$-x^2 + 8x - 16 = 0$; $(x-4)^2 = 0$; $x = 4$ ✓

$y = -4 + 8 = 4$; R$(4\,;\,4)$ ✓

$f(4) = -\frac{1}{2}(4-2)^2 + 8 = 6$ ✓

$4 = 6 + t$, so $t = -2$ ✓

Alternative: $g'(x) = -1$ gives $x = 4$ and $y = 4$, then continue as above.

6.11.1 (3)

At A: $-x^2 - 2x + 8 = 0$; $(x+4)(x-2) = 0$; A$(-4\,;\,0)$ ✓

$m = \frac{8-0}{0+4} = 2$ ✓

$y - 0 = 2(x+4)$, so $y = 2x + 8$ ✓

Substituting A or C into $y = 2x + c$ is also accepted for the last mark.

6.11.2 (4)

EH $= f(x) - g(x) = (-x^2 - 2x + 8) - (2x + 8)$ ✓
$= -x^2 - 4x$ ✓
Maximum when $x = \dfrac{-b}{2a} = \dfrac{4}{-2} = -2$ ✓
EH $= -(-2)^2 - 4(-2) = 4$ units ✓

Answer only: 1 out of 4 marks.

> **Memo trap:** Using the turning point of f ($x = -1$). The longest vertical distance is where the gradients of f and g are equal.

6.11.3 (5)

$k'(x) = -2x - 2m - 2$ ✓
$k'(x) = 2$ ✓
$x = -m - 2$ ✓
$k(-m - 2) = f(-2) = 8$ and $g(-m - 2) = 2(-m - 2) + 8$; equating: $8 = -2m + 4$ ✓
$m = -2$ ✓

Alternative: $k(x) = g(x)$ gives $x^2 + (2m + 4)x + m^2 + 2m = 0$; $\Delta = 0$ gives $m = -2$ (same 5 marks).

Answer only: 0 out of 5 marks.

6.12.1 Read off the graph: g is above f between the vertical asymptote and D ✓ (2)
$3 < x < 4$ ✓

Accepted: $x \in (3\,;\,4)$. $x = 3$ is not included.

6.12.2 $f(x) = -6$ at $x = 4$ (D) and, by symmetry about $x = 3$, at $x = 2$ ✓ (2)
$2 < x < 4$ ✓

6.12.3 P is the meeting point of the asymptotes of g: P$(3\,;\,-8)$, so $f(x) = a(x - 3)^2 - 8$ ✓ (3)
Substitute D$(4\,;\,-6)$: $-6 = a(4 - 3)^2 - 8$, so $a = 2$ ✓
$f(x) = 2(x - 3)^2 - 8 = 2x^2 - 12x + 10$ ✓

6.12.4 (6)

$f(x) = 0$: $2x^2 - 12x + 10 = 0$ ✓
$x^2 - 6x + 5 = 0$; $(x - 1)(x - 5) = 0$ ✓
M$(1\,;\,0)$ (the smaller root) ✓
$g(x) = 0$: $\dfrac{2}{x - 3} - 8 = 0$ ✓
$x - 3 = \dfrac{1}{4}$, so T$\left(\dfrac{13}{4}\,;\,0\right)$ ✓
MT $= \dfrac{13}{4} - 1 = \dfrac{9}{4} = 2{,}25$ units ✓

6.12.5 (3)

$f'(x) = 4x - 12$ ✓
$m = f'(4) = 4$ ✓
$-6 = 4(4) + c$, so $c = -22$ and $y = 4x - 22$ ✓

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QUESTION 7

Finance, Growth And Decay

7.1.1 (2)

$A = 36\,500(1 + 0{,}064)^4$ ✓
Cost $= \text{R}46\,779{,}91$ ✓

7.1.2 (3)

$i = \dfrac{0{,}056}{4}$ and $n = 18 \times 4 = 72$ ✓

$A = 8\,000\left(1 + \dfrac{0{,}056}{4}\right)^{72}$ ✓

$A = \text{R}21\,768{,}21$ ✓

7.1.3 (3)

$A = P(1 - i)^n$ ✓

$A = 640\,000(1 - 0{,}086)^6$ ✓

$A = \text{R}373\,127{,}41$ ✓

7.1.4 (3)

$118\,636{,}75 = 320\,000(1 - 0{,}18)^n$ ✓

$(0{,}82)^n = 0{,}370740$ and $n = \dfrac{\log 0{,}370740}{\log 0{,}82}$ ✓

$n = 5$ years ✓

Using logs to any base is accepted. Trial and error with the right answer: full marks.

7.1.5 (2)

$0 = 15\,000(1 - 0{,}125n)$ ✓

$n = 8$ years ✓

7.1.6 (2)

$0{,}4P = P(1 - 5i)$ ✓

$i = 0{,}12$, so the rate is 12% p.a. ✓

Alternative: $\dfrac{60\%}{5} = 12\%$ (method + answer).

7.2.1 (5)

$17\,364{,}22 = 14\,000\left(1 + \dfrac{r}{12}\right)^{36}$ ✓✓

$\left(1 + \dfrac{r}{12}\right)^{36} = 1{,}240301$

$1 + \dfrac{r}{12} = \sqrt[36]{1{,}240301} = 1{,}006000$ ✓

$\dfrac{r}{12} = 0{,}006000$ ✓

$r = 7{,}2\%$ ✓

Marks: $\dfrac{r}{12}$ (1); $n = 36$ (1); substitution (1); 36th root (1); answer (1).

7.2.2 (2)

$1 + i_{\text{eff}} = \left(1 + \dfrac{7{,}2}{1200}\right)^{12}$ ✓

$i_{\text{eff}} = 7{,}44\%$ ✓

Unrounded r gives $7{,}44\%$; accept either.

7.2.3 (4)

$n = 5$ ✓

$10\,285 = 9\,500\left(1 + \dfrac{m}{400}\right)^5$ ✓

$1 + \dfrac{m}{400} = \sqrt[5]{\dfrac{10\,285}{9\,500}} = 1{,}016006$ ✓

$m = 6{,}40\%$ ✓

7.2.4 (4)

$88\,838{,}74 = 60\,000\left(1 + \dfrac{0{,}072}{4}\right)^{4n}$ ✓✓

$(1{,}018)^{4n} = 1{,}480646$

$4n = \dfrac{\log 1{,}480646}{\log 1{,}018} = 22$ ✓

$n = 5{,}5$ years ✓

Marks: $\dfrac{0{,}072}{4}$ (1); substitution (1); logs (1); answer (1).

7.2.5 (5)

Zinzi: $A = 8\,000(1 + 0{,}091 \times 5)$ ✓ $= \text{R}11\,640{,}00$

With bonus: $11\,640{,}00 \times 1{,}03 = \text{R}11\,989{,}20$ ✓

Ben: $A = 8\,000\left(1 + \dfrac{0{,}084}{4}\right)^{20}$ ✓ $= \text{R}12\,122{,}85$ ✓

Ben's investment is worth more ✓

7.3.1 (4)

$n = 109$ ✓

$i = \dfrac{0{,}084}{12}$ ✓

$$F = \frac{750\left[\left(1 + \frac{0{,}084}{12}\right)^{109} - 1\right]}{\frac{0{,}084}{12}}$$ ✓

$F = \text{R}122\,036{,}51$ ✓

7.3.2 (2)

$A = 122\,036{,}51\left(1 + \dfrac{0{,}084}{12}\right)^{12}$ ✓

$A = \text{R}132\,691{,}60$ ✓

7.3.3 (4)

$i = \dfrac{0{,}072}{12}$ ✓

$n = 12$ ✓

$$F = \frac{1\,200\left[\left(1 + \frac{0{,}072}{12}\right)^{12} - 1\right]}{\frac{0{,}072}{12}} = \text{R}14\,884{,}83$$ ✓

R14 884,83 is less than R15 500, so she does NOT have enough (she is short by R615,17) ✓

7.3.4 (4)

$i = \dfrac{0{,}093}{12}$ and $n = 60$ ✓

$$380\,000 = \frac{x\left[\left(1 + \frac{0{,}093}{12}\right)^{60} - 1\right]}{\frac{0{,}093}{12}}$$ ✓✓

$x = \text{R}4\,998{,}62$ ✓

Marks: i (1); n (1); substitution into correct formula (1); answer (1).

7.3.5 (4)

$i = \dfrac{0{,}064}{4}$ and $n = 24$ deposits ✓

$$F = \frac{1\,800\left[\left(1 + \frac{0{,}064}{4}\right)^{24} - 1\right]}{\frac{0{,}064}{4}}$$ ✓

$\ldots \times \left(1 + \dfrac{0{,}064}{4}\right)$ ✓

$F = \text{R}52\,999{,}72$ ✓

Equivalent route: $F = \dfrac{1\,800\left[(1+i)^{25} - 1\right]}{i} - 1\,800 = \text{R}52\,999{,}72$ (same marks).

> **Memo trap:** Using $n = 24$ without multiplying by $(1 + i)$ treats the deposits as made at the END of each quarter.

7.3.6 (3)

$i = \dfrac{0{,}076}{4}$ and $n = 16$ ✓

$$F = \frac{12\,000\left[\left(1 + \frac{0{,}076}{4}\right)^{16} - 1\right]}{\frac{0{,}076}{4}}$$ ✓

$F = \text{R}221\,942{,}80$ ✓

7.3.7 (3)

$$221\,942{,}80 - 80\,000\left(1 + \frac{0{,}076}{4}\right)^{4}$$ ✓✓

$= \text{R}135\,687{,}31$ ✓

Alternative: amount at the end of 3 years $= \text{R}80\,043{,}05$, then grown for one more year, and the four deposits of the last year added (same marks).

> **Memo trap:** Subtracting only R80 000 forgets that the withdrawn money would have earned interest for the last year.

7.4.1 (2)

$$A = 320\,000\left(1 + \frac{0{,}132}{12}\right)^{3}$$ ✓

$A = \text{R}330\,676{,}59$ ✓

7.4.2 (4)

$i = \dfrac{0{,}132}{12}$ ✓

$n = 57$ ✓

$$330\,676{,}59 = \frac{x\left[1 - \left(1 + \frac{0{,}132}{12}\right)^{-57}\right]}{\frac{0{,}132}{12}}$$ ✓

$x = \text{R}7\,839{,}72$ ✓

Using the rounded balance R330 676,59 gives R7 839,72; accept.

> **Memo trap:** 1 July 2021 to 1 March 2026 is 56 months, but there are 57 payments because both the first and the last are counted.

7.4.3 (5)

$$330\,676{,}59 = \frac{9\,000\left[1 - \left(1 + \frac{0{,}132}{12}\right)^{-n}\right]}{\frac{0{,}132}{12}}$$ ✓✓

$$\left(1 + \frac{0{,}132}{12}\right)^{-n} = 0{,}595840$$

$$n = -\frac{\log 0{,}595840}{\log 1{,}011} = 47{,}33$$ ✓

48 payments are needed ✓

$57 - 48 = 9$ months sooner ✓

7.4.4 (4)

Balance after 47 payments $= 330\,676{,}59\left(1+\frac{0{,}132}{12}\right)^{47}$ ✓

$$-\frac{9\,000\left[\left(1+\frac{0{,}132}{12}\right)^{47}-1\right]}{\frac{0{,}132}{12}}$$ ✓

$= \text{R}2\,945{,}36$

Final instalment $= 2\,945{,}36\left(1+\frac{0{,}132}{12}\right)$ ✓

$= \text{R}2\,977{,}76$ ✓

Alternative: the balance after 48 payments of R9 000 would be −R6 022,24 (overpaid), so the last instalment is R9 000 − R6 022,24 = R2 977,76.

Method 2: balance after 47 payments $= \frac{9\,000\left[1-\left(1+\frac{0{,}132}{12}\right)^{-0{,}3297}\right]}{\frac{0{,}132}{12}} = \text{R}2\,945{,}36$ using the fractional part of n as the number of payments still owing, then $\times\left(1+\frac{0{,}132}{12}\right)$. Same marks.

7.5.1 (4)

$i = \frac{0{,}126}{12}$ ✓

$n = 48$ ✓

$$184\,000 = \frac{x\left[1-\left(1+\frac{0{,}126}{12}\right)^{-48}\right]}{\frac{0{,}126}{12}}$$ ✓

$x = \text{R}4\,899{,}81$ ✓

7.5.2 (6)

Total paid in year 1: $4\,899{,}81 \times 12 = \text{R}58\,797{,}72$ ✓

Balance after 12 months $= 184\,000\left(1+\frac{0{,}126}{12}\right)^{12}$ ✓

$$-\frac{4\,899{,}81\left[\left(1+\frac{0{,}126}{12}\right)^{12}-1\right]}{\frac{0{,}126}{12}}$$ ✓

$= \text{R}146\,255{,}87$ ✓

Capital repaid $= 184\,000 - 146\,255{,}87 = \text{R}37\,744{,}13$ ✓

Interest $= 58\,797{,}72 - 37\,744{,}13 = \text{R}21\,053{,}59$ ✓

Alternative for the balance: $P = \frac{4\,899{,}81\left[1-(1+i)^{-36}\right]}{i} = \text{R}146\,255{,}96$; the interest then differs by a few cents. Accept. The same method is asked when the loan comes from a given instalment (value of the loan first) or when the balance comes from an earlier part.

7.6.1 (4)

$i = \dfrac{0{,}099}{12}$ ✓

$n = 240$ ✓

$$1\,350\,000 = \frac{x\left[1 - \left(1 + \frac{0{,}099}{12}\right)^{-240}\right]}{\frac{0{,}099}{12}}$$ ✓

$x = \text{R}12\,938{,}48$ ✓

7.6.2 (5)

$$\text{Balance} = 1\,350\,000\left(1 + \frac{0{,}099}{12}\right)^{150}$$ ✓ $$- \frac{12\,938{,}48\left[\left(1 + \frac{0{,}099}{12}\right)^{150} - 1\right]}{\frac{0{,}099}{12}}$$ ✓

$= 4\,629\,865{,}45 - 3\,810\,233{,}32$ ✓✓

$= \text{R}819\,632{,}13$ ✓

Method 2: $P = \dfrac{12\,938{,}48\left[1 - \left(1 + \frac{0{,}099}{12}\right)^{-90}\right]}{\frac{0{,}099}{12}} = \text{R}819\,633{,}85$ with $n = 90$ remaining payments. Full marks (the answers differ by a few rand because of the rounded instalment).

7.7 (3)

$n = 84$ ✓

$$P = \frac{1\,800\left[1 - \left(1 + \frac{0{,}109}{12}\right)^{-84}\right]}{\frac{0{,}109}{12}}$$ ✓

$P = \text{R}105\,448{,}76$ ✓

7.8 (3)

Loan $= \text{R}1\,200\,000$

Payments made: $17 \times 12 = 204$, so 36 payments remain

$$P = \frac{12\,714{,}57\left[1 - \left(1 + \frac{0{,}114}{12}\right)^{-36}\right]}{\frac{0{,}114}{12}}$$ ✓✓

$P = \text{R}386\,126{,}99$ ✓

OR: $n = 204$ in both formulae: $1\,200\,000(1+i)^{204} - \dfrac{12\,714{,}57[(1+i)^{204} - 1]}{i} = \text{R}386\,127{,}81$.

Marks: n (1); substitution (1); answer (1).

7.9.1 (1)

R272 000 ✓

$80\% \times 340\,000$

7.9.2 Amount owing one month before the first repayment (4 months of interest): (5)

$$A = 272\,000\left(1 + \frac{0{,}126}{12}\right)^{4} \checkmark = \text{R}283\,605{,}19 \checkmark$$

Payments are made in months 5 to 60, so $n = 56$ ✓

$$283\,605{,}19 = \frac{x\left[1 - \left(1 + \frac{0{,}126}{12}\right)^{-56}\right]}{\frac{0{,}126}{12}} \checkmark$$

$$x = \text{R}6\,724{,}18 \checkmark$$

> **Memo trap:** Payments are made in months 5 to 60, which is 56 payments. Using $n = 60$ or $n = 55$ loses the last two marks.

7.10.1 Amount paid $= 3\,123{,}08 \times 72 = \text{R}224\,861{,}76$ ✓ (2)

Interest $= 224\,861{,}76 - 150\,000 = \text{R}74\,861{,}76$ ✓

7.10.2 On 1 August 2024, 36 payments have been made, so $n = 36$ payments remain ✓ (7)

$$i = \frac{0{,}144}{12} \checkmark$$

$$\text{Balance} = \frac{3\,123{,}08\left[1 - \left(1 + \frac{0{,}144}{12}\right)^{-36}\right]}{\frac{0{,}144}{12}} \checkmark = \text{R}90\,860{,}56 \checkmark$$

After the extra payment: $90\,860{,}56 - 25\,000 = \text{R}65\,860{,}56$ ✓

$$65\,860{,}56 = \frac{3\,123{,}08\left[1 - \left(1 + \frac{0{,}144}{12}\right)^{-n}\right]}{\frac{0{,}144}{12}} \text{ gives } (1+i)^{-n} = 0{,}746940 \checkmark$$

$n = 24{,}46$, so 25 more payments; $36 - 25 = 11$ months earlier ✓

Alternative: $L = \frac{x[1-(1+i)^{-n}]}{i} + 20\,000(1+i)^{-36}$ solved for n (the number of months after starting payments); then subtract from 72 (same marks).

7.11.1 (5)

$$i = \frac{0{,}096}{12} \checkmark$$

$$640\,000 = \frac{7\,200\left[1 - \left(1 + \frac{0{,}096}{12}\right)^{-n}\right]}{\frac{0{,}096}{12}} \checkmark$$

$$1 - \left(1 + \frac{0{,}096}{12}\right)^{-n} = 0{,}711111 \checkmark$$

$$n = -\frac{\log 0{,}288889}{\log\left(1 + \frac{0{,}096}{12}\right)} = 155{,}83 \checkmark$$

156 payments ✓

7.11.2 Extra paid each month $= 7\,200 - 6\,007{,}49 = \text{R}1\,192{,}51$ ✓ (4)

$n = 96$ ✓

$$F = \frac{1\,192{,}51\left[\left(1 + \frac{0{,}096}{12}\right)^{96} - 1\right]}{\frac{0{,}096}{12}}$$ ✓

$F = \text{R}171\,255{,}56$ ✓

7.12 $i = \frac{0{,}102}{12}$ ✓ (5)

$$1\,200\,000 = \frac{15\,000\left[1 - \left(1 + \frac{0{,}102}{12}\right)^{-n}\right]}{\frac{0{,}102}{12}}$$ ✓

$$\left(1 + \frac{0{,}102}{12}\right)^{-n} = 0{,}320000$$

$$-n = \frac{\log 0{,}320000}{\log\left(1 + \frac{0{,}102}{12}\right)}$$ ✓

$n = 134{,}62$ ✓

He can make 134 full withdrawals ✓ (round DOWN: a part withdrawal is not R15 000)

Marks: i; substitution; logs; n; whole number of withdrawals.

7.13 (5)

Balance on 30 April 2023: $A = 680\,000\left(1 + \frac{0{,}084}{12}\right)^{3} = \text{R}694\,380{,}19$ ✓

$$694\,380{,}19 = \frac{9\,500\left[1 - \left(1 + \frac{0{,}084}{12}\right)^{-n}\right]}{\frac{0{,}084}{12}}$$ ✓

$\left(1 + \frac{0{,}084}{12}\right)^{-n} = 0{,}488351$, so $-n = \log_{1{,}007} 0{,}488351$ ✓

$n = 102{,}75$ payments ✓

103 payments, the last one a smaller payment; the loan is repaid 3 + 103 = 106 months after it was granted ✓

The first payment is in month 4, so payment number k is made in month $k + 3$.

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8.1.1 (5)

$f(x+h) = 3(x+h)^2 - 4 = 3x^2 + 6xh + 3h^2 - 4$ ✓

$f'(x) = \lim_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$ ✓

$= \lim_{h\to 0} \dfrac{3x^2 + 6xh + 3h^2 - 4 - (3x^2 - 4)}{h} = \lim_{h\to 0} \dfrac{6xh + 3h^2}{h}$ ✓

$= \lim_{h\to 0} \dfrac{h(6x + 3h)}{h} = \lim_{h\to 0}(6x + 3h)$ ✓

$= 6x$ ✓

Limit notation on every line until h is replaced by 0. Leaving it out, or writing $f'(x) = \ldots$ with h still in the expression, loses 1 mark for notation (the penalty is applied once in the question).

8.1.2 (6)

$f'(x) = \lim_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$ ✓

$= \lim_{h\to 0} \dfrac{3(x+h)^2 - 2(x+h) - (3x^2 - 2x)}{h}$ ✓

$= \lim_{h\to 0} \dfrac{3x^2 + 6xh + 3h^2 - 2x - 2h - (3x^2 - 2x)}{h}$ ✓

$= \lim_{h\to 0} \dfrac{6xh + 3h^2 - 2h}{h}$ ✓

$= \lim_{h\to 0} \dfrac{h(6x + 3h - 2)}{h} = \lim_{h\to 0}(6x + 3h - 2)$ ✓

$= 6x - 2$ ✓

Limit notation on every line (see 8.1.1).

In 2021 and 2022 the question was worth 5 marks: the formula and the substitution shared one mark.

8.1.3 (4)

$f(x+h) = 8 - 5(x+h) = -5x - 5h + 8$ ✓

$f'(x) = \lim_{h\to 0} \dfrac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \dfrac{-5x - 5h + 8 - (8 - 5x)}{h}$ ✓

$= \lim_{h\to 0} \dfrac{-5h}{h}$ ✓

$= \lim_{h\to 0}(-5) = -5$ ✓

Limit notation on every line (see 8.1.1).

8.1.4 (2)

$g(x) = \dfrac{1}{x}$ ✓

$a = 3$ ✓

The limit has the form $\lim_{h\to 0} \dfrac{g(a+h) - g(a)}{h}$. Compare: $g(a+h) = \dfrac{1}{3+h}$ and $g(a) = \dfrac{1}{3}$.

8.2.1 (3)

$\dfrac{dy}{dx} = 10x^4$ ✓ $-21x^2$ ✓ $+4$ ✓

One mark for each term.

8.2.2 (2)

$= 9 - 8x$ ✓✓

One mark for each term.

8.2.3 (2)

$D_x\left[4x^2 + 3x - 10\right]$ ✓

$= 8x + 3$ ✓

Expand first. The product rule is not needed.

8.2.4 (4)

$y = x^{\frac{5}{2}} - 6x^{-2}$ ✓✓

$\dfrac{dy}{dx} = \dfrac{5}{2}x^{\frac{3}{2}} + 12x^{-3}$ ✓✓

Also accepted: $\dfrac{5}{2}\sqrt{x^3} + \dfrac{12}{x^3}$. One mark for each rewritten term and one for each derivative.

8.2.5 (4)

$y = 4x^{-1} + x^{\frac{3}{2}} - \frac{1}{3}\pi$ ✓✓

$\frac{dy}{dx} = -4x^{-2} + \frac{3}{2}x^{\frac{1}{2}}$ ✓✓

π is a constant, so $\frac{1}{3}\pi$ differentiates to 0.

> **Memo trap:** Treating π as a variable (for example $-\frac{1}{3}$ as an extra term) loses the last mark.

8.2.6 (3)

$y = 3x^6 + x^{\frac{3}{4}}$ ✓

$\frac{dy}{dx} = 18x^5$ ✓ $+\frac{3}{4}x^{-\frac{1}{4}}$ ✓

8.2.7 (4)

$D_x\left[-\frac{1}{4}x^{\frac{1}{2}} + \frac{1}{8}x^{-3}\right]$ ✓✓

$= -\frac{1}{8}x^{-\frac{1}{2}} - \frac{3}{8}x^{-4}$ ✓✓

Also accepted: $-\frac{1}{8\sqrt{x}} - \frac{3}{8x^4}$.

8.2.8 (3)

$D_x\left[10x^{\frac{3}{5}} + 4x^{-3}\right]$ ✓

$= 6x^{-\frac{2}{5}} - 12x^{-4}$ ✓✓

One mark for the rational exponent, then one for each derivative.

8.2.9 (4)

$g(x) = 3x^{-4} - x^{\frac{5}{3}}$ ✓✓

$g'(x) = -12x^{-5} - \frac{5}{3}x^{\frac{2}{3}}$ ✓✓

8.2.10 (4)

$y(x-2) = 3x(x-2)$ ✓✓

$y = 3x$ (because $x \neq 2$) ✓

$\frac{dy}{dx} = 3$ ✓

One mark for each factorised side. Make y the subject first.

8.2.11 (4)

$f(x) = \frac{(5x-2)(5x+2)}{2(5x-2)}$ ✓✓

$= \frac{5x+2}{2} = \frac{5}{2}x + 1$ ✓

$f'(x) = \frac{5}{2}$ ✓

One mark for each factorised expression. After cancelling, the function is a straight line, so the derivative is its gradient.

8.2.12 (4)

$y = 4x^3 + 3x^{-2}$ ✓✓

$\frac{dy}{dx} = 12x^2 - 6x^{-3}$ ✓✓

Divide each term by x^2 first. The quotient rule is not needed.

8.3.1 (1)

$\frac{dy}{dx} = 3ax^2$ ✓

8.3.2 (2)

$\frac{dy}{da} = x^3 - 5$ ✓✓

Now x is the constant and a is the variable. One mark for x^3, one for -5.

8.4.1 (5)

$f'(x) = 3x^2 + 2ax + b$ ✓

$f'(2) = -5$: $12 + 4a + b = -5$, so $4a + b = -17$ ✓

$f(2) = g(2) = -2$: $8 + 4a + 2b + 12 = -2$, so $4a + 2b = -22$ ✓

$b = -5$ ✓ and $a = -3$ ✓

The tangent and the curve share the point $(2; -2)$ and have the same gradient there.

8.4.2 (4)

$b = 4 + \frac{8}{4} = 6$ ✓

$\frac{dy}{dx} = 1 - \frac{8}{x^2}$ ✓

$m_{\tan} = 1 - \frac{8}{16} = \frac{1}{2}$, so $m_{\perp} = -2$ ✓

$y - 6 = -2(x - 4)$, so $y = -2x + 14$ ✓

8.4.3 (3)

$f'(x) = 3x^2 - 10x + 4$

$m = f'(3) = 1$ ✓

$f(3) = -4$ ✓

$y - (-4) = 1(x - 3)$, so $y = x - 7$ ✓

8.5.1 (3)

$f'(x) = -3x^2 + 6x + 9 > 0$ ✓

$-3(x - 3)(x + 1) > 0$, so the critical values are $x = -1$ and $x = 3$ ✓

$-1 < x < 3$ ✓

A parabola that opens downwards is positive BETWEEN its roots.

> **Memo trap:** Writing $x < -1$ or $x > 3$: the sign of the quadratic is the wrong way round.

8.5.2 (4)

$g'(x) = 3ax^2 + 12x + b$ and $g''(x) = 6ax + 12$ ✓

$g''(-2) = 0$: $-12a + 12 = 0$ ✓

$a = 1$ ✓

$g''(x) > 0$: $x > -2$ ✓

Alternative: the minimum gradient is at a point of inflection, so $x = -2$ (1), the cubic has $a > 0$ (1) and it is concave up for $x > -2$ (1).

8.6.1 (5)

$f(x + h) = -3(x + h)^2 = -3x^2 - 6xh - 3h^2$ ✓

$f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}$ ✓

$= \lim_{h \to 0} \frac{-3x^2 - 6xh - 3h^2 - (-3x^2)}{h} = \lim_{h \to 0} \frac{-6xh - 3h^2}{h}$ ✓

$= \lim_{h \to 0} \frac{h(-6x - 3h)}{h} = \lim_{h \to 0}(-6x - 3h)$ ✓

$= -6x$ ✓

Limit notation on every line (see 8.1.1).

8.6.2 (1)

$x \geq 0$ (or $x \leq 0$) ✓

Either restriction is accepted.

8.6.3 (3)

Swap x and y: $x = -3y^2$ ✓

$y^2 = -\frac{x}{3}$, so $y = \pm\sqrt{-\frac{x}{3}}$ ✓

$y = \sqrt{-\frac{x}{3}}$, $x \leq 0$ ✓

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9.1.1 (3)

$f'(x) = 0$ ✓

$6x^2 - 6x - 12 = 0$, so $6(x-2)(x+1) = 0$ ✓

$x = 2$ or $x = -1$ ✓

9.1.2 (3)

$f''(x) = 12x - 6$ ✓

$f''(x) < 0$: $12x - 6 < 0$ ✓

$x < \frac{1}{2}$ ✓

9.1.3 (2)

$f'(x) > 0$ for $x < -1$ ✓ or $x > 2$ ✓

Interval notation $(-\infty; -1) \cup (2; \infty)$ is also accepted. The memo of the original paper also accepted $\leq$ and $\geq$.

9.1.4 (5)

$d = 5$ ✓

$f'(x) = 3ax^2 + 2bx + c = 6x^2 - 6x - 12$ ✓

$3a = 6$, so $a = 2$ ✓

$2b = -6$, so $b = -3$ ✓

$c = -12$ ✓

$f(x) = 2x^3 - 3x^2 - 12x + 5$

Alternative: integrate $f'(x)$ to get $f(x) = 2x^3 - 3x^2 - 12x + d$ (3 marks), then $d = 5$ (1 mark) and the answer (1 mark).

9.2.1 (5)

$f(x) = x^3 - 3x^2 + 4$ ✓

$f'(x) = 3x^2 - 6x$ ✓

$f''(x) = 6x - 6$ ✓

$f''(x) = 0$: $6x - 6 = 0$, so $x = 1$ ✓

$f''(0) = -6 < 0$ and $f''(2) = 6 > 0$: f'' changes sign at $x = 1$, so there is a point of inflection ✓

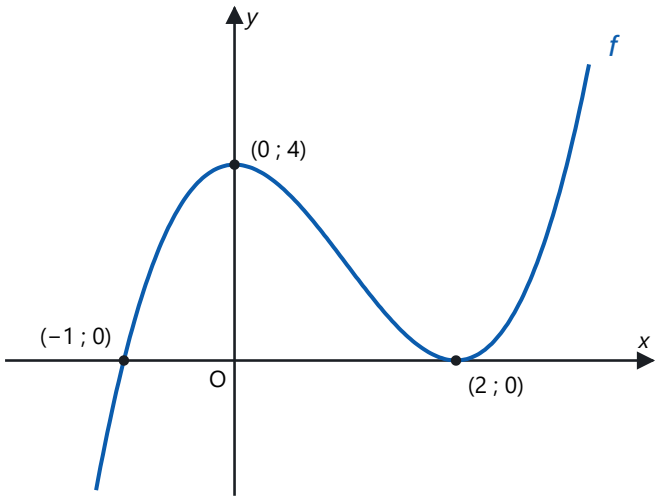
9.2.2 (4)

Shape: positive cubic ✓

x-intercept $(-1; 0)$ ✓

Local maximum $(0; 4)$, which is also the y-intercept ✓

Local minimum $(2; 0)$ on the x-axis (the graph touches the axis) ✓



9.2.3 f is concave up for $x > 1$ (from $f''(x) > 0$) ✓ (2)

$-f$ is concave down for $x > 1$ ✓

Reflecting in the x-axis turns a concave-up part into a concave-down part.

9.2.4 f is moved 2 units left and 3 units down, so $(0; 4)$ moves to $(-2; 1)$ ✓✓ (2)

One mark for each coordinate.

9.2.5 No ✓ (2)

Between $x = 0$ and $x = 2$ the graph of f is decreasing, so the gradient at $x = 1$ is negative. (Also $f'(1) = -3$.) ✓

9.3.1 P is a turning point on the x-axis, so g has a repeated root: $g(x) = (x + 4)(x - p)^2$ ✓ (4)

$g(0) = 4$: $4p^2 = 4$ ✓

$p = 1$ (because P lies to the right of the y-axis) ✓

$g(x) = (x + 4)(x^2 - 2x + 1) = x^3 + 2x^2 - 7x + 4$, so $b = 2$, $c = -7$ and $d = 4$ ✓

9.3.2 $g'(x) = 3x^2 + 4x - 7$ ✓ (5)

$g'(x) = 0$ ✓

$(3x + 7)(x - 1) = 0$ ✓

$R\left(-\frac{7}{3}; \frac{500}{27}\right)$ or $R(-2{,}33; 18{,}52)$ ✓

$P(1; 0)$ ✓

9.3.3 $g''(x) = 6x + 4$ ✓ (3)

$g''(0) = 4$ ✓

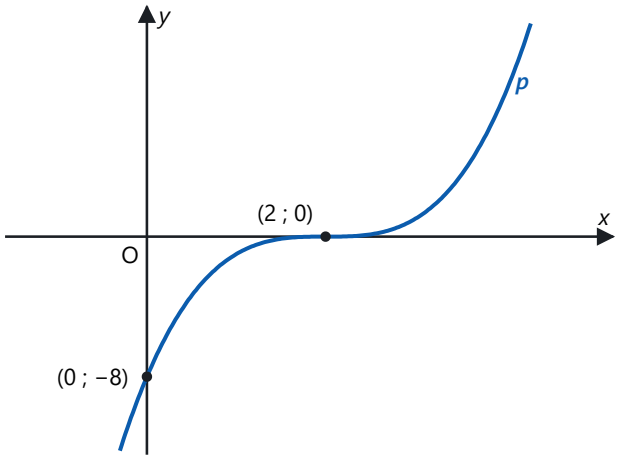
$g''(0) > 0$, so g is concave up at $(0; 4)$ ✓

Alternative: $6x + 4 = 0$ gives $x = -\frac{2}{3}$ as the point of inflection, and $x = 0$ lies to its right, so g is concave up (3 marks).

9.3.4 y-intercept $(0; -8)$ ✓ (3)

Stationary point and point of inflection at $(2; 0)$ ✓

Concave down for $x < 2$ and concave up for $x > 2$ (increasing cubic) ✓



9.4.1 $f'(x) = 6x^2$ ✓ (3)

$2x^3 = 6x^2$, so $2x^2(x - 3) = 0$

$x = 0$ ✓ or $x = 3$ ✓

9.4.2 f and f' ✓ (1)

9.4.3 For f: $(0;0)$ is a point of inflection ✓ (2)
For f': $(0;0)$ is a turning point (a minimum) ✓

9.4.4 $f''(x) = 12x$ ✓ (3)
Distance $= f''(1) - f'(1) = 12(1) - 6(1)^2$ ✓
$= 6$ ✓

9.4.5 $2x^3 - 6x^2 < 0$ ✓ (4)
$2x^2(x-3) < 0$ ✓
$x < 3$ ✓ and $x \neq 0$ ✓
$2x^2 > 0$ for every $x \neq 0$, so only the sign of $x - 3$ matters. At $x = 0$ the expression equals 0, not less than 0.

9.5.1 $-2 < x < 1$ ✓✓ (2)

9.5.2 $x = \dfrac{-2+1}{2} = -\dfrac{1}{2}$ ✓✓ (2)
The point of inflection lies halfway between the turning points. Answer only: full marks.

9.5.3 $x > -\dfrac{1}{2}$ ✓✓ (2)
Answer only: full marks.

9.5.4 $g'(x) = 3ax^2 + 2bx + c = -6x^2 - 6x + 12$ ✓ (4)
$3a = -6$, so $a = -2$ ✓
$2b = -6$, so $b = -3$ ✓
$c = 12$, so $g(x) = -2x^3 - 3x^2 + 12x$ ✓

9.5.5 The maximum gradient is at the turning point of g', that is at $x = -\dfrac{1}{2}$ ✓ (5)
$g'\left(-\dfrac{1}{2}\right) = \dfrac{27}{2} = 13{,}5$ ✓
$g\left(-\dfrac{1}{2}\right) = -\dfrac{13}{2} = -6{,}5$ ✓
$y - (-6{,}5) = 13{,}5\,(x - (-0{,}5))$ ✓
$y = 13{,}5x + 0{,}25$ ✓

9.6.1 $h'(x) = 3ax^2 + 2bx$ ✓ (5)
$h'(3) = 0$: $27a + 6b = 0$, so $9a + 2b = 0$ ✓
$h(3) = 27$: $27a + 9b = 27$, so $3a + b = 3$ ✓
Solve simultaneously: $9a + 2b = 0$ and $6a + 2b = 6$ ✓
$3a = -6$, so $a = -2$ and $b = 3 - 3(-2) = 9$ ✓

9.6.2 $-2x^3 + 9x^2 = 0$ ✓ (3)
$x^2(-2x + 9) = 0$ ✓
$x = 0$ or $x = \dfrac{9}{2}$, so $A\left(\dfrac{9}{2};0\right)$ ✓

9.6.3 $0 < x < 3$ (critical values ✓, notation ✓) (2)
Interval notation $x \in (0;3)$ is accepted.

9.6.4 $x > \dfrac{3}{2}$ (critical value ✓, notation ✓) (2)
The point of inflection is at $h''(x) = -12x + 18 = 0$, that is $x = \dfrac{3}{2}$.

9.6.5 The equation is $h(x+2)=k$: the graph of h moves 2 units to the left (3)

New y-intercept: $h(2)=20$ ✓

The local maximum value is 27, so $k<27$ ✓

$20<k<27$ ✓

The turning points move to $(-2;0)$ and $(1;27)$. For two positive roots the line $y=k$ must cut the graph above the new y-intercept (20) and below the local maximum (27).

9.7.1 $f'(x)=m(x+2)\left(x-\frac{2}{3}\right)$ ✓ (6)

Substitute $(0;4)$: $4=m(2)\left(-\frac{2}{3}\right)$ ✓

$m=-3$ ✓

$f'(x)=-3\left(x^2+\frac{4}{3}x-\frac{4}{3}\right)$ ✓

$n=-4$ ✓

$k=4$ ✓

Alternative: $k=4$ from R; then $4m-2n+4=0$ and $\frac{4}{9}m+\frac{2}{3}n+4=0$, solved simultaneously.

9.7.2 $f'(x)=0$ gives $x=-2$ or $x=\frac{2}{3}$ ✓ (3)

$f(-2)=0$, so the turning point is $(-2;0)$ ✓

$f\left(\frac{2}{3}\right)=\frac{256}{27}$, so the turning point is $\left(\frac{2}{3};\frac{256}{27}\right)$, approximately $(0{,}67;9{,}48)$ ✓

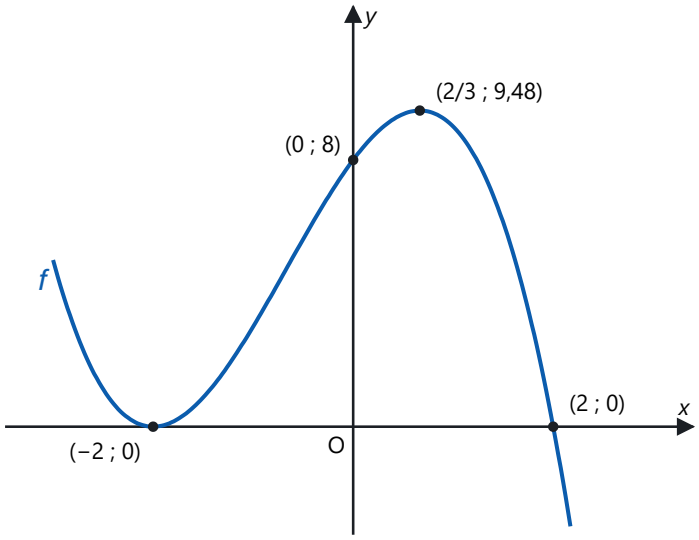
9.7.3 Shape: negative cubic (rises to the left, falls to the right) ✓ (5)

y-intercept $(0;8)$ ✓

$-x^3-2x^2+4x+8=-(x-2)(x+2)^2=0$, so the x-intercept $(2;0)$ ✓

Local minimum $(-2;0)$ on the x-axis (the graph touches the axis) ✓

Local maximum $\left(\frac{2}{3};9{,}48\right)$ ✓



9.7.4 (1)

$a = \dfrac{-2 + \frac{2}{3}}{2} = -\frac{2}{3}$ ✓

E and W are symmetrical about the axis of symmetry of f'. Alternative: $f''(a) = -6a - 4 = 0$ gives $a = -\frac{2}{3}$.

9.7.5 (2)

The maximum value of f' is $f'\left(-\frac{2}{3}\right) = \frac{16}{3}$ ✓, so $b < \frac{16}{3}$ ✓

The tangents at E and W always meet ABOVE the turning point of f'. If D lies below the turning point, no such tangents exist.

9.8.1 (4)

$f'(x) = -6x^2 + 24x - 18$ ✓

$f'(x) = 0$: $-6(x - 1)(x - 3) = 0$ ✓

$x = 1$ or $x = 3$ ✓

$f(1) = -8$ and $f(3) = 0$: turning points $(1; -8)$ and $(3; 0)$ ✓

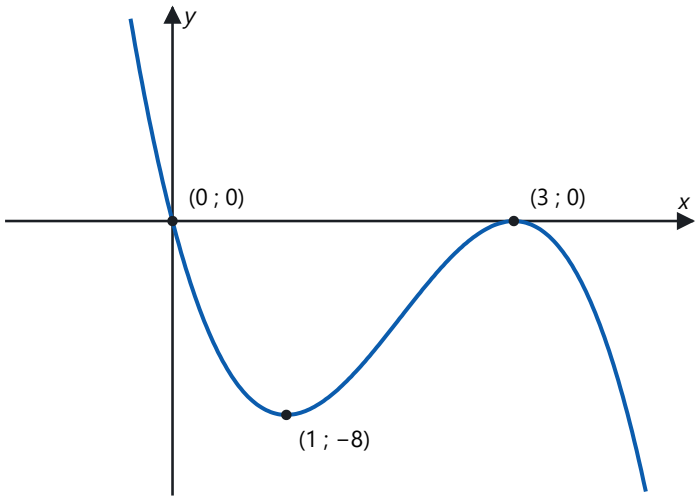
9.8.2 (4)

Shape: negative cubic ✓

Intercepts: $(0; 0)$ (this is also the y-intercept) and $(3; 0)$ ✓

Local minimum $(1; -8)$ ✓

Local maximum $(3; 0)$ on the x-axis (the graph touches the axis) ✓



9.8.3 (2)

$-8 < k < 0$ ✓✓

The line $y = k$ must cut the graph between the y-values of the two turning points.

9.8.4 (6)

$f''(x) = -12x + 24$ ✓

$f''(x) = 0$ ✓

$x = 2$ ✓

$f(2) = -4$ ✓

$f'(2) = 6$ ✓

$y - (-4) = 6(x - 2)$, so $g(x) = 6x - 16$ ✓

9.8.5 (2)

$\tan\theta = 6$ ✓

$\theta = 80{,}54°$ ✓

9.9.1 (2)

$2 < x < 4$ ✓✓

9.9.2 (2)

$x = 2$ ✓ and $x = 4$ ✓

The turning points are where $f'(x) = 0$.

9.9.3 Point of inflection at $x = \dfrac{2+4}{2} = 3$ ✓ (2)

$x > 3$ ✓

9.9.4 $-10 < k < -6$ ✓✓ (2)

Moving the graph down by $|k|$: the local minimum (6) must go below the x-axis and the local maximum (10) must stay above it.

9.10.1 $f'(x) = 3x^2 - 10x - 8$ ✓ (4)

$f'(x) = 0$: $(3x+2)(x-4) = 0$ ✓

$x = -\dfrac{2}{3}$ or $x = 4$; E is the local minimum, so $x = 4$ ✓

$f(4) = -36$, so $E(4;-36)$ ✓

9.10.2 $f''(x) = 6x - 10$ ✓ (3)

$f''(x) < 0$: $6x - 10 < 0$ ✓

$x < \dfrac{5}{3}$ ✓

Equivalent route: x is the midpoint of the turning points, $\dfrac{-\frac{2}{3}+4}{2} = \dfrac{5}{3}$.

9.10.3 x-intercepts: -2, 1 and 6 ✓✓ (4)

$f(x) > 0$ and $f''(x) < 0$: $-2 < x < 1$ ✓

$f(x) < 0$ and $f''(x) > 0$: $\dfrac{5}{3} < x < 6$ ✓

The product is negative when the two factors have opposite signs.

9.10.4 $f'(x) = -8$: $3x^2 - 10x - 8 = -8$ ✓ (6)

$3x^2 - 10x = 0$, so $x(3x - 10) = 0$ ✓

$x = 0$ or $x = \dfrac{10}{3}$ ✓

At $x = 0$: $t = f(0) + 8(0) = 12$ ✓

At $x = \dfrac{10}{3}$: $t = f\left(\dfrac{10}{3}\right) + 8\left(\dfrac{10}{3}\right) = -\dfrac{176}{27}$, that is $t \approx -6{,}52$ ✓

$-\dfrac{176}{27} < t < 12$ ✓

The line is parallel to the tangents at $x = 0$ and $x = \dfrac{10}{3}$. At those two values of t the line is a tangent and meets f at only 2 points, so the endpoints are NOT included.

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QUESTION 10

Calculus: Optimisation And Rates Of Change

10.1.1 $N(4) = -(4)^3 + 6(4)^2 + 36(4)$ ✓ (2)

$= 176$ thousand bacteria ✓

10.1.2 $N'(t) = -3t^2 + 12t + 36$ ✓ (3)

$N'(3) = -3(3)^2 + 12(3) + 36$ ✓

$= 45$ thousand bacteria per hour ✓

10.1.3 $N''(t) = -6t + 12$ ✓ (3)

$N''(t) = 0$ ✓

$t = 2$ hours ✓

The rate of change is largest where the second derivative is zero (the graph of N' has its turning point).

10.2.1 $h(0) = (0-8)(-14) = 112$ cm ✓ (1)

10.2.2 $h(t) = 0$: $t = 8$ ✓ (3)
For $-2t^2 + 2t - 14 = 0$: $\Delta = 2^2 - 4(-2)(-14) = -108 < 0$ ✓
No real roots, so the gecko reached the floor only ONCE ✓

10.2.3 $h(t) = -2t^3 + 18t^2 - 30t + 112$ ✓ (4)
$h'(t) = -6t^2 + 36t - 30 = 0$ ✓
$t^2 - 6t + 5 = 0$; $(t-1)(t-5) = 0$; $t = 1$ or $t = 5$ ✓
$h(5) = 162$ cm ✓
$h(1) = 98$ cm is a local minimum, so $t = 5$ gives the maximum height.

10.3.1 $s''(t) = -6t + 30 = 0$ ✓ (3)
$t = 5$ ✓
$s'(5) = 75$ km/h ✓

10.3.2 The train stops when $s'(t) = 0$: $-3t(t-10) = 0$ ✓ (5)
$t = 0$ or $t = 10$ ✓
$s(t) = at^3 + bt^2$ with $3a = -3$ and $2b = 30$ ✓
$s(t) = -t^3 + 15t^2$ ✓
$s(10) = 500$ km ✓
Marks: factors; values of t; a and b (by working back from the derivative); substitution of $t = 10$; answer.
The constant term is 0 because $s(0) = 0$.

10.4 $P(x; x^2+1)$ and $B(0; 2{,}5)$: $PB^2 = (x-0)^2 + (x^2+1-2{,}5)^2$ ✓ (7)
$= x^4 - 2x^2 + 2{,}25$ ✓
$\dfrac{d(PB^2)}{dx} = 4x^3 - 4x$ ✓
$4x^3 - 4x = 0$ ✓
$4x(x^2-1) = 0$, so $x = 1$ (since $x > 0$) ✓
$PB^2 = 1 - 2 + 2{,}25 = 1{,}25$ ✓
$PB = \sqrt{1{,}25} = 1{,}12$ units ✓
Alternative: the tangent at P has gradient $2x$ and must be perpendicular to BP: $2x \times \dfrac{x^2+1-2{,}5}{x-0} = -1$, giving $x = 1$ and $PB = \dfrac{\sqrt{5}}{2} = 1{,}12$.

10.5 Any point on f: $(x; x^2)$ ✓ (8)
$d^2 = (x-6)^2 + (x^2-3)^2$ ✓
$= x^4 - 5x^2 - 12x + 45$ ✓
$\dfrac{d(d^2)}{dx} = 4x^3 - 10x - 12$ ✓
$= 0$ ✓
$(x-2)(4x^2+8x+6) = 0$; the quadratic has $\Delta = -32 < 0$, so $x = 2$ ✓
$d^2 = 17$ ✓
$d = \sqrt{17} = 4{,}12$ units ✓
Alternative (perpendicular to the tangent): $2x \times \dfrac{x^2-3}{x-6} = -1$ gives $2x^3 - 5x - 6 = 0$, so $x = 2$ and $d = \sqrt{(2-6)^2 + (4-3)^2} = \sqrt{17}$. Answer only: full marks.

10.6.1 $AH : HG = 2 : 1$ ✓ (1)

$DE = FC = \frac{2}{3}BC$, and $\triangle ADE \sim \triangle ABC$.

10.6.2 (5)

$HG = \frac{1}{3}t$ ✓

$FC = \frac{2}{3}(6 - t)$ ✓

$A(t) = \frac{2}{3}(6 - t) \times \frac{1}{3}t = \frac{4}{3}t - \frac{2}{9}t^2$ ✓

$A'(t) = \frac{4}{3} - \frac{4}{9}t = 0$ ✓

$t = 3$ ✓

10.7.1 (4)

Total surface area $= 2lw + 2wh + 2lh$ ✓

With $l = 2w$: $= 4w^2 + 2wh + 4wh = 4w^2 + 6wh$ ✓

Top and bottom: $12(4w^2) = 48w^2$ ✓

Sides: $8(6wh) = 48wh$ ✓

$\text{Cost} = 48w^2 + 48wh$

10.7.2 (6)

$32 = 2w^2h$, so $h = \frac{16}{w^2}$ ✓

$C(w) = 48w^2 + 48w\left(\frac{16}{w^2}\right)$ ✓

$= 48w^2 + 768w^{-1}$ ✓

$C'(w) = 96w - 768w^{-2}$ ✓

$96w - \frac{768}{w^2} = 0$ ✓

$w^3 = 8$, so $w = 2$ m ✓

Then $h = 4$ m and the minimum cost is R576.

10.8 (7)

Time $= \frac{150}{x}$ hours ✓

Wages: $40 \times \frac{150}{x} = \frac{6000}{x}$ ✓

Petrol: $150\left(1{,}4 + \frac{x}{250}\right)$ ✓

$C(x) = \frac{6000}{x} + 210 + 0{,}6x$ ✓

$C'(x) = -\frac{6000}{x^2} + 0{,}6$ ✓

$-\frac{6000}{x^2} + 0{,}6 = 0$ ✓

$x^2 = 10\,000$, so $x = 100$ km/h ✓

The minimum cost is R330,00.

10.9.1 (3)

$AB = \frac{360}{x}$ ✓

Width of notice $= (x + 10)$ ✓

Height of notice $= \left(\frac{360}{x} + 4\right)$ ✓

$A(x) = (x + 10)\left(\frac{360}{x} + 4\right) = 360 + 4x + \frac{3600}{x} + 40 = \frac{3600}{x} + 4x + 400$

10.9.2 $A(x) = 3600x^{-1} + 4x + 400$ ✓ (3)

$A'(x) = -\dfrac{3600}{x^2} + 4 = 0$ ✓

$x^2 = 900$, so $x = 30$ cm ✓

10.10.1 $2x + 2h = 36$, so $h = 18 - x$ ✓ (3)

$2\pi r = x$, so $r = \dfrac{x}{2\pi}$ ✓

$V = \pi r^2 h = \pi\left(\dfrac{x}{2\pi}\right)^2 (18 - x) = \dfrac{x^2(18 - x)}{4\pi} = \dfrac{18x^2 - x^3}{4\pi}$ ✓

10.10.2 $V'(x) = \dfrac{36x - 3x^2}{4\pi}$ ✓ (3)

$V'(x) = 0$: $3x(12 - x) = 0$ ✓

$x \neq 0$, so $x = 12$ units ✓

Then $h = 6$ units, which satisfies $x > h$.

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QUESTION 11

Probability

11.1.1 $P(A \text{ or } B) = P(A) + P(B)$ ✓ (3)

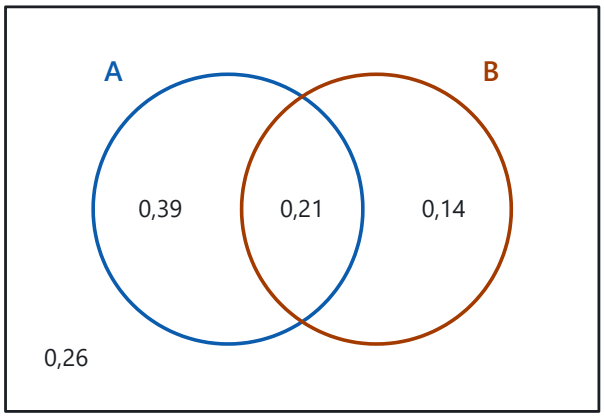
$0{,}7 = 0{,}4 + y$ ✓

$y = 0{,}30$ ✓

11.2.1 $P(A \text{ and } B) = P(A) \times P(B) = 0{,}6 \times 0{,}35 = 0{,}21$ ✓ (3)

Only A: $0{,}39$; only B: $0{,}14$ ✓

Outside both circles: $0{,}26$ ✓



11.2.2 $P(A \text{ or not } B) = P(A) + P(\text{not } B) - P(A \text{ and not } B) = 0{,}6 + 0{,}65 - 0{,}39$ ✓ (2)

$= 0{,}86$ ✓

Alternative: $1 - P(\text{only } B) = 1 - 0{,}14 = 0{,}86$.

11.3.1 No, because $P(A \text{ and } B) \neq 0$ ✓ (1)

11.3.2 $P(A \text{ and } B) = P(A) \times P(B)$ ✓ (4)

$P(B) = 0{,}24 + 0{,}16 = 0{,}40$ ✓

$0{,}24 = P(A) \times 0{,}40$, so $P(A) = 0{,}60$ ✓

$P(\text{only } A) = 0{,}60 - 0{,}24 = 0{,}36$ ✓

11.3.3 $P(\text{not } A \text{ or not } B) = 1 - P(A \text{ and } B)$ ✓ (2)
$= 1 - 0{,}24 = 0{,}76$ ✓
Venn route: $P(\text{only } A) + P(\text{only } B) + P(\text{neither}) = 0{,}36 + 0{,}16 + 0{,}24 = 0{,}76$.

11.4.1 $P(A \text{ and } B) = \frac{2}{5} \times \frac{5}{8} = \frac{1}{4}$ ✓✓ (2)

11.4.2 $P(A \text{ or } B) = \frac{2}{5} + \frac{5}{8} - \frac{1}{4}$ ✓ (2)
$= \frac{31}{40}$ ✓
Venn alternative: $P(\text{only } A) + P(A \text{ and } B) + P(\text{only } B) = \frac{3}{20} + \frac{1}{4} + \frac{3}{8} = \frac{31}{40}$.

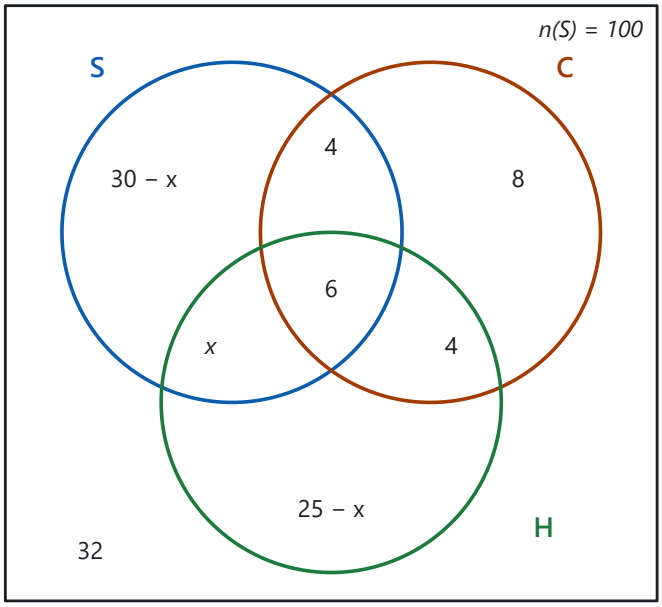
11.5.1 $y = 1 - 0{,}85 = 0{,}15$ ✓ (1)

11.5.2 $x = 0{,}85 - 0{,}77 = 0{,}08$ ✓ (1)
0,77 is the sum of the other six regions: 0,15 + 0,20 + 0,05 + 0,10 + 0,15 + 0,12.

11.5.3 $P = 0{,}10 + 0{,}15 + 0{,}12 + 0{,}08$ ✓ (2)
$= 0{,}45$ ✓

11.5.4 $P(B) = 0{,}20 + 0{,}10 + 0{,}12 + 0{,}08 = 0{,}50$ ✓ (5)
$P(C) = 0{,}05 + 0{,}15 + 0{,}12 + 0{,}08 = 0{,}40$ ✓
$P(B \text{ and } C) = 0{,}12 + 0{,}08 = 0{,}20$ ✓
$P(B) \times P(C) = 0{,}50 \times 0{,}40 = 0{,}20$ ✓
$P(B \text{ and } C) = P(B) \times P(C)$, so B and C ARE independent ✓

11.6.1 Eight regions filled in correctly with the values shown ✓✓✓✓ (4)
1 mark for 2 or 3 regions correct, 2 marks for 4 or 5 correct, 3 marks for 6 or 7 correct, 4 marks for all 8 correct.



11.6.2 $(30 - x) + 8 + (25 - x) + 4 + 4 + x + 6 + 32 = 100$ ✓ (2)
$109 - x = 100$, so $x = 9$ ✓

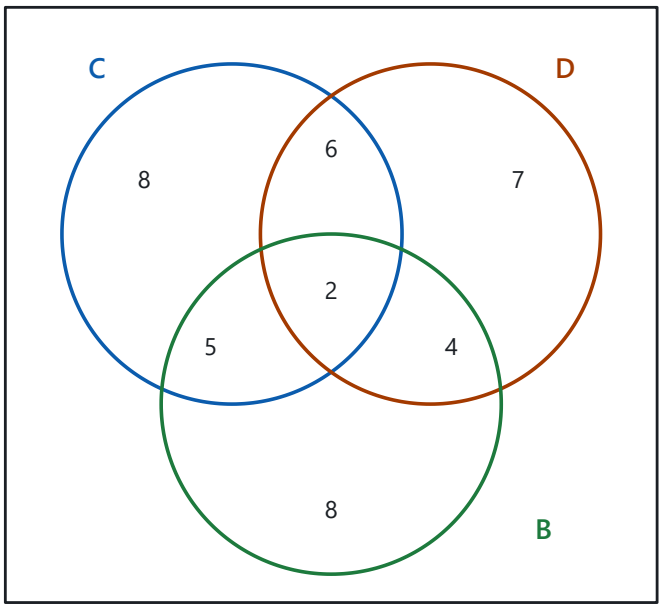
11.6.3 $P = \frac{21 + 8 + 16}{100}$ ✓ (2)
$= \frac{45}{100} = 0{,}45$ ✓

11.7.1 2 in the centre ✓ (3)

6 ; 5 and 4 in the overlaps ✓

8 ; 7 and 8 for the single clubs ✓

Check: C = 8+6+5+2 = 21 ; D = 7+6+4+2 = 19 ; B = 8+5+4+2 = 19. Total = 40 learners.



11.7.2 $P = \frac{6+5+4+2}{40}$ ✓ (2)

$= \frac{17}{40} = 0{,}425$ ✓

11.7.3 $P(C) \times P(D) = \frac{21}{40} \times \frac{19}{40} = 0{,}249$ ✓✓ (4)

$P(C \text{ and } D) = \frac{8}{40} = 0{,}20$ ✓

$P(C) \times P(D) \neq P(C \text{ and } D)$, so the events are NOT independent ✓

11.8.1 $a + 22 = 40$, so $a = 18$ ✓ (2)

$b = 72 + 48 = 120$ ✓

11.8.2 No ✓ (2)

$P(\text{boy and does not play}) = \frac{18}{160} \neq 0$, so the events can occur together ✓

11.8.3 $P(\text{plays a sport}) = \frac{120}{160} = \frac{3}{4} = 0{,}75$ ✓✓ (2)

11.8.4 $P(\text{girl and does not play}) = \frac{22}{160} = \frac{11}{80} = 0{,}1375$ ✓✓ (2)

11.9.1 $P(M \text{ and } W) = P(M) \times P(W)$ ✓ (3)

$\frac{40}{250} = \frac{100}{250} \times \frac{e}{250}$ ✓

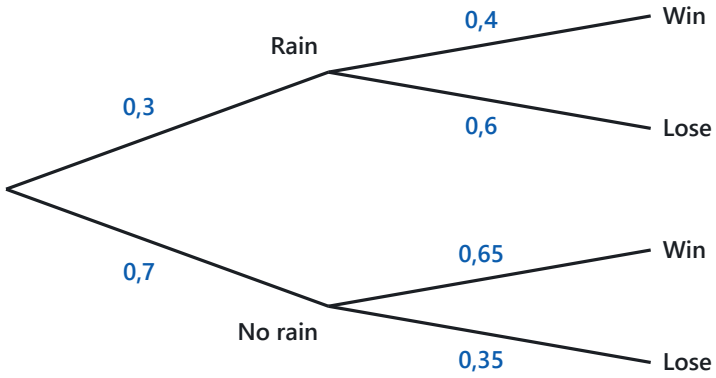
$P(M) = \frac{60+40}{250} = \frac{100}{250}$, so $e = 100$ ✓

11.9.2 $d = 250 - 100 = 150$ ✓ (3)

$b = 150 - 60 = 90$ ✓

$P(\text{female and taxi}) = \frac{90}{250} = \frac{9}{25} = 0{,}36$ ✓

11.10.1 First branches: rain 0,3 and no rain 0,7 ✓ (3)
From rain: win 0,4 and lose 0,6 ✓
From no rain: win 0,65 and lose 0,35 ✓



11.10.2 $P(\text{rain and lose}) = 0,3 \times 0,6 = 0,18$ ✓ (3)
$P(\text{no rain and lose}) = 0,7 \times 0,35 = 0,245$ ✓
$P(\text{not win}) = 0,18 + 0,245 = 0,425$ ✓

11.11 Let x be the probability of buying iced tea on a day that is not hot, so $2x$ on a hot day. (4)
$\frac{3}{5}(2x)$ ✓
$+\frac{2}{5}(x)$ ✓
$= \frac{2}{5}$ ✓, so $8x = 2$ and $x = \frac{1}{4}$
Number of customers $= 160 \times \frac{1}{4} = 40$ ✓

11.12.1 (2)
$P = \frac{5}{25}$ ✓
$= \frac{1}{5} = 0,20$ ✓

11.12.2 Consecutive pairs: Mon-Tue, Tue-Wed, Wed-Thu, Thu-Fri, each in 2 orders: 4×2 ✓ (3)
Total outcomes $= 5 \times 5 = 25$ ✓
$P = \frac{8}{25} = 0,32$ ✓

11.13.1 $P = 0,4 \times 0,4 = 0,16$ ✓✓ (2)

11.13.2 Exactly two hits: $0,4 \times 0,4 \times 0,6 = 0,096$ in each of 3 orders ✓ (3)
Three hits: $0,4 \times 0,4 \times 0,4 = 0,064$ ✓
$P = 3(0,096) + 0,064 = 0,352$ ✓

11.13.3 $P = 0,4 + (0,6)(0,6)(0,4) + \ldots$ ✓✓ (3)
$a = 0,4$; $r = 0,36$
$P = \frac{0,4}{1 - 0,36} = 0,625$ ✓
Accept $\frac{5}{8}$.

11.14 Let the number of bags with a mystery gift be x; total number of bags $= 3x$ ✓ (6)

$\frac{x}{3x}$ or $\frac{1}{3}$ ✓

$\frac{x-1}{3x-1}$ ✓

$\frac{1}{3} \times \frac{x-1}{3x-1} = \frac{4}{39}$ ✓✓

$39x - 39 = 36x - 12$, so $x = 9$ ✓

There are 9 bags with a mystery gift inside (27 bags in total).

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QUESTION 12

Counting Principle

12.1 First digit 7: $1 \times 6 \times 5 \times 3 = 90$ ✓✓ (5)

First digit 6 or 8: $2 \times 6 \times 5 \times 4 = 240$ ✓✓

Total $= 90 + 240 = 330$ ✓

Alternative: last digit 1, 3 or 5 and first digit 6, 7 or 8: $3 \times 3 \times 6 \times 5 = 270$; last digit 7 and first digit 6 or 8: $1 \times 2 \times 6 \times 5 = 60$; total 330.

12.2.1 $6 \times 6 \times 10 \times 9$ ✓✓ (3)

$= 3\,240$ ✓

12.2.2 Five digits: $6 \times 6 \times 10 \times 9 \times 8 \times 7 \times 6 = 1\,088\,640$ ✓✓ (3)

$1\,088\,640 \geq 1\,000\,000$, so FIVE digits ✓

Four digits give only $6 \times 6 \times 10 \times 9 \times 8 \times 7 = 181\,440$ codes, which is not enough.

12.3.1 $8^4 = 4\,096$ ✓✓ (2)

12.3.2 $8 \times 7 \times 6 \times 5 = 1\,680$ ✓✓ (2)

Also accepted: $\frac{8!}{4!}$.

12.3.3 $4 \times 8 \times 1 = 32$ ✓✓✓ (3)

Hundreds digit: 3, 4, 5 or 6 (4 choices). Tens digit: any of the 8 digits. Units digit: must be 5, since 0 is not available.

12.4.1 $10^9 = 1\,000\,000\,000$ ✓✓ (2)

12.4.2 Region code 9×10 and branch code $8 \times 10 \times 8$ ✓ (3)

Serial number $10 \times 10 \times 10 \times 2$ ✓

Total $= (9 \times 10)(8 \times 10 \times 8)(10 \times 10 \times 10 \times 2) = 115\,200\,000$ ✓

This is $1{,}152 \times 10^8$.

12.4.3 $P = \frac{115\,200\,000}{1\,000\,000\,000}$ ✓ (2)

$= 0{,}1152 = 11{,}52\%$ ✓

12.5.1 $26 \times 26 \times 10 \times 10 = 67\,600$ ✓✓ (2)

12.5.2 $21 \times 21 \times 9 \times 5$ ✓✓✓✓ (4)

$= 19\,845$

First letter: 26 − 4 − 1 = 21. Second letter: 22 allowed letters, one already used = 21. Last digit: 0, 2, 4, 6 or 8 = 5. Third character (digit): 10 − 1 = 9. One mark for each correct factor.

12.5.3 $25 \times 25 \times 9 \times 5 = 28\,125$ ✓ (2)

$\frac{28\,125 - 19\,845}{19\,845} \times 100 = 41{,}72\%$ ✓

12.6 Seat Red: $6 \times 1 \times 4$; seat Black: $6 \times 1 \times 4$ ✓✓ (6)
Seat Tan: $6 \times 1 \times 5$; seat Grey: $6 \times 1 \times 5$ ✓✓
Number of variations $= 24 + 24 + 30 + 30 = 108$ ✓
Floor space needed $= 108 \times 5 = 540\ \text{m}^2 < 600\ \text{m}^2$, so the display IS possible ✓
Alternative: $6 \times 4 \times 5 = 120$; $120 - 6 \times 2 = 108$ (the 2 colour pairs that are the same, for each body style), then 540 m² < 600 m².

12.7.1 $P = \frac{3}{10}$ ✓ (1)

12.7.2 $10! = 3\,628\,800$ ✓✓ (2)

12.7.3 First and last positions: 3×4 ✓ (4)
Language books together: $3! = 6$ ✓
The other books and the block: $6!$ ✓
$P = \frac{3 \times 4 \times 6! \times 3!}{10!} = \frac{51\,840}{3\,628\,800} = \frac{1}{70}$ ✓
The 3 language books count as ONE block, so the 2 other mathematics books, the 3 other science books and the block make 6 items.

12.8.1 The last digit has 3 choices (3 ; 5 or 7) ✓ (3)
The other digits are chosen from the 6 remaining allowed digits ✓
$3 \times 6 \times 5 \times 4 = 360$ ✓

12.8.2 First digit 5 or 7 (last digit has 2 choices): $2 \times 1 \times 4 \times 2 = 16$ ✓ (5)
First digit 6 or 8 (last digit has 3 choices): $2 \times 1 \times 4 \times 3 = 24$ ✓
Number of favourable codes $= 16 + 24 = 40$ ✓
$P = \frac{40}{360}$ ✓
$= \frac{1}{9} = 0{,}11$ ✓
The third digit can be any of the 4 allowed digits left over after the first, second and last digits have been chosen.

12.9.1 $9! = 362\,880$ ✓ (1)

12.9.2 The 2 oldest learners can take 4 pairs of positions with exactly 4 learners between them: (1 and 6), (2 and 7), (3 and 8), (4 and 9) (4)
For each pair: $2 \times 7!$ ✓
Favourable $= 4(2 \times 7!) = 40\,320$ ✓✓
$P = \frac{40\,320}{9!} = \frac{1}{9}$ ✓

12.10.1 Treat Dumi and Chipo as one block: $6! = 720$ ✓✓ (2)

12.10.2 Dumi in position 1: Chipo in position 4, 5, 6 or 7 (4 ways); Dumi in 2: 3 ways; Dumi in 3: 2 ways; Dumi in 4: 1 way (4)
Favourable outcomes $= 5!(4 + 3 + 2 + 1) = 5!(10)$ ✓✓
Total outcomes $= 7!$ ✓
$P = \frac{5!(10)}{7!} = \frac{5}{21} = 0{,}24$ ✓

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